

Role of Posterior Medial Thalamus in the Modulation of Striatal Circuitry and Choice Behavior

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eLife Assessment

Yonk and colleagues provide a **valuable**, timely, and in-depth study showcasing the role of thalamostriatal inputs in learning and action selection. After characterizing the synaptic properties of these inputs onto different striatal cell types in vitro, they provide **solid** evidence that posterior medial thalamic nucleus (POm) terminals in striatum are activated during reward expectation and arousal. The overall function of this pathway and the degree to which results are confounded by viral contamination of surrounding nuclei and movements remain open questions.

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Abstract

Summary

The posterior medial (POm) thalamus is heavily interconnected with sensory and motor circuitry and is likely involved in behavioral modulation and sensorimotor integration. POm provides axonal projections to the dorsal striatum, a hotspot of sensorimotor processing, yet the role of POm-striatal projections has remained undetermined. Using optogenetics with slice electrophysiology, we found that POm provides robust synaptic input to direct and indirect pathway striatal spiny projection neurons (D1- and D2-SPNs, respectively) and parvalbumin-expressing fast spiking interneurons (PVs). During the performance of a whisker-based tactile discrimination task, POm-striatal projections displayed learning-related activation correlating with anticipatory, but not reward-related, pupil dilation. Inhibition of POm-striatal axons across learning caused slower reaction times and an increase in the number of training sessions for expert performance. Our data indicate that POm-striatal inputs provide a behaviorally relevant arousal-related signal, which may prime striatal circuitry for efficient integration of subsequent choice-related inputs.

Introduction

The process of sensorimotor learning is underpinned by sensory perception and motor control.¹ In the mouse whisker system, tactile sensations are acquired via active sensor (e.g., whisker) movement to obtain relevant environmental information and subsequent processing by the well-characterized primary somatosensory barrel cortex (S1) circuitry.^{1,2} This whisker-related information is transmitted from the periphery to S1 via two thalamic nuclei, ventral posterior medial (VPM) and posterior medial (POm), constituting the lemniscal and paralemniscal pathways, respectively.^{3,4,5,6,7,8,9,10} VPM reliably encodes fast-whisking components including self-motion and tactile information.^{7,11,12,13,14} Conversely, POm encodes phase-related whisking activity with relatively lower magnitude responses and higher response failure rates.^{7,12,13,14,15,16,17,18} Recent work has highlighted two behavior-related aspects of POm function: (1) activation during changes in behavioral state, especially related to sensory and nociceptive processing,^{7,11,12,13,17,18,19,20,21} and (2) driving learning-related plasticity at its cortical synapses.^{22,23,24,25,26}

POm receives a plethora of inputs including glutamatergic (S1, primary motor cortex (M1), secondary somatosensory cortex, superior colliculus, and spinal trigeminal interpolaris),^{15,27,28,29,30,31,32,33,34} GABAergic (ventral zona incerta, anterior pretectal nucleus, and thalamic reticular nucleus),^{16,35,36,37,38,39,40,41,42} and cholinergic (pedunculopontine and laterodorsal tegmental nuclei).^{43,44,45,46} Further, the stereotypical POm-cortical projection terminates in S1, specifically layers (L)1 and L5A,^{4,47,48,49} and has been studied in the context of driving cortical plasticity/perceptual learning.^{22,23,24,25,26,50,51} In addition to its cortical projection, POm axons pass through and collateralize with terminal synaptic boutons in both thalamic reticular nucleus and posterior dorsolateral striatum (pDLS) as they ascend towards cortex.^{49,52,53,54,55,56,57,58} Here, we focus on the POm-striatal projection as striatal circuitry modulation may have powerful effects on sensorimotor integration and behavior.⁵⁹ However, POm's influence over striatal microcircuitry and behavioral performance is unresolved.

The striatum is the predominant input nucleus of the basal ganglia and is predominantly composed of GABAergic spiny projection neurons (SPNs) expressing either D1 or D2 dopamine receptors,^{60,61,62} but it also contains a rich diversity of interneurons, such as parvalbumin-expressing (PV) fast-spiking interneurons that exert robust modulatory control over SPN output.^{63,64,65,66,67} Within this microcircuitry, the dorsal striatum integrates widespread convergent cortical and thalamic inputs that constitute part of the force driving normal striatal functioning.^{68,69,70,71,72,73,74,75,76} Notably, functionally-related cortical (S1 and M1) and thalamic (POm) inputs converge within shared striatal subregions.^{73,74,77} For example, M1 and S1 are heavily interconnected via reciprocal L2/3 and L5 connections,^{78,79,80} and their projections overlap within dorsal striatum and even onto the same neuron.^{73,81,82,83,84,85,86} While some studies treat striatal inputs as a uniform entity,^{87,88} they have been shown to differ anatomically,^{73,74,76} functionally,^{89,90} and behaviorally.^{89,91,92} Thus, the specific cortical and thalamic origin of striatal inputs likely has significance for understanding how the striatal circuitry integrates sensorimotor information to modulate behavior.

The most prominent thalamostriatal modulation occurs via parafascicular (Pf) thalamus.^{90,93,94,95,96,97,98,99,100,101} Pf is implicated in regulating action flexibility^{102,103} and contributing to the initiation and execution of learned sequences of movements^{101,104} through its robust innervation of striatal cholinergic interneurons.^{95,87,88,90} Conversely, despite direct comparisons to Pf,⁵⁶ POm's functional

innervation pattern and subsequent influence over the striatal microcircuitry and choice behavior is undetermined.⁵⁹ Here, we used *ex vivo* whole-cell recordings of identified D1-SPNs, D2-SPNs, and PV interneurons to assess the functional connectivity of POM-striatal projections, and *in vivo* fiber photometry and photoinactivation to identify the contribution of POM-striatal axonal activity on sensory-guided behavioral performance and learning.

Results

POM Equally Innervates Striatal Cell Types

With Faster Latency In PV Interneurons

We stereotactically injected pAAV-ChR2-EYFP unilaterally in POM, permitting channelrhodopsin (ChR2) expression in its thalamostriatal terminals to investigate the relative synaptic strength of POM inputs onto three identified striatal neurons (D1-SPNs, D2-SPNs, and PV interneurons; **Figure 1A-B**). The injection site was confirmed by verifying the stereotypical POM-cortical projection pattern (S1 L1 and L5a; **Figure 1C**).^{4,25,26,47,49} In acute *ex vivo* brain slices, neurons were targeted for patch clamp recordings within pDLS (AP from bregma = -0.34 to -1.22) corresponding with the POM-striatal axonal projection field (**Figure S1**).^{56,105} Striatal neurons were identified and targeted by crossing their respective Cre-recombinase mouse lines with tdTomato-expressing reporter mice and validating their intrinsic electrophysiological properties in response to hyperpolarizing and depolarizing current steps (**Figure S1A-D, G-L**; see Methods).^{64,67,106} Whole-cell current-clamp recordings were performed without inhibitory synaptic blockers to resemble natural physiological responses to optogenetic activation of POM inputs.⁸⁹ After breaking in, patched cells were subjected to a standard set of protocols: (1) hyperpolarizing and depolarizing current steps to define intrinsic firing properties and optogenetic; and (2) single pulse (SP), (3) paired-pulse ratio (PPR), and (4) train stimulation to measure synaptic responses.

Optogenetic activation of POM terminals readily elicited depolarizing postsynaptic potentials (PSP) in all targeted cell types (**Figure 1D**). Responses to SP stimulation resulted in relatively equal PSP amplitudes for D1-SPNs (7.05±0.75mV), D2-SPNs (8.79±1.67mV), PVs (10.83±1.91mV), and neighboring unlabeled cells that we termed putative SPNs, based on their intrinsic firing properties (5.54±1.04mV) ($F_{(4,51)} = 2.455$, $p = 0.4835$; **Figure 1E**). Three PV interneurons and one D1-SPN exhibited action potentials to SP stimulation and were excluded from further analysis. A small but significant correlation was observed between PSP amplitude and increasing distance from the injection site ($r^2 = 0.08493$, $p = 0.0293$, $n = 59$ cells; **Figure S1E-F**). The latency to maximum PSP amplitude was significantly shorter in PVs (7.29±0.32ms, $n = 17$ cells from 7 mice) than D1-SPNs (11.05±0.45ms, $n = 20$ cells from 6 mice), D2-SPNs (10.68±0.56ms, $n = 11$ cells from 5 mice), and putative SPNs (10.81±0.92ms, $n = 7$ cells from 4 mice) ($F_{(4,51)} = 29.78$, $p < 0.0001$, PV vs. D1 $p < 0.0001$, PV vs. D2 $p = 0.0003$, PV vs. SPN, $p = 0.0056$; **Figure 1F**). In a subset of recordings, identified and unidentified cells within the same field of view on the same slice were patched sequentially to control for injection site variability. D1- and D2-SPNs did not differ in PSP amplitude (9.55±2.83mV in D1-SPNs vs. 8.48±2.15mV in D2-SPNs, $p = 0.6406$, $n = 8$ pairs, $N = 5$) or latency (10.13±0.66ms in D1-SPNs vs. 10.92±0.56ms in D2-SPNs, $p = 0.3254$, $n = 8$ pairs, $N = 5$; **Figure S1M-O**). In contrast, sequentially patched PV and SPNs did not significantly differ in maximum PSP amplitude (10.26±3.05mV in PVs vs. 4.76±0.97mV in SPNs, $p = 0.2500$, $n = 9$ pairs, $N = 7$), but PVs had faster latency (6.85±0.42ms in PVs vs. 10.75±0.87ms in SPNs, $p = 0.0006$, $n = 9$ pairs, $N = 7$; **Figure S1P-R**), validating the population results. Strikingly, we found relatively equal PSP amplitudes in all recorded cell types within pDLS, indicating that POM provides robust and unbiased synaptic input to all targeted striatal cells.

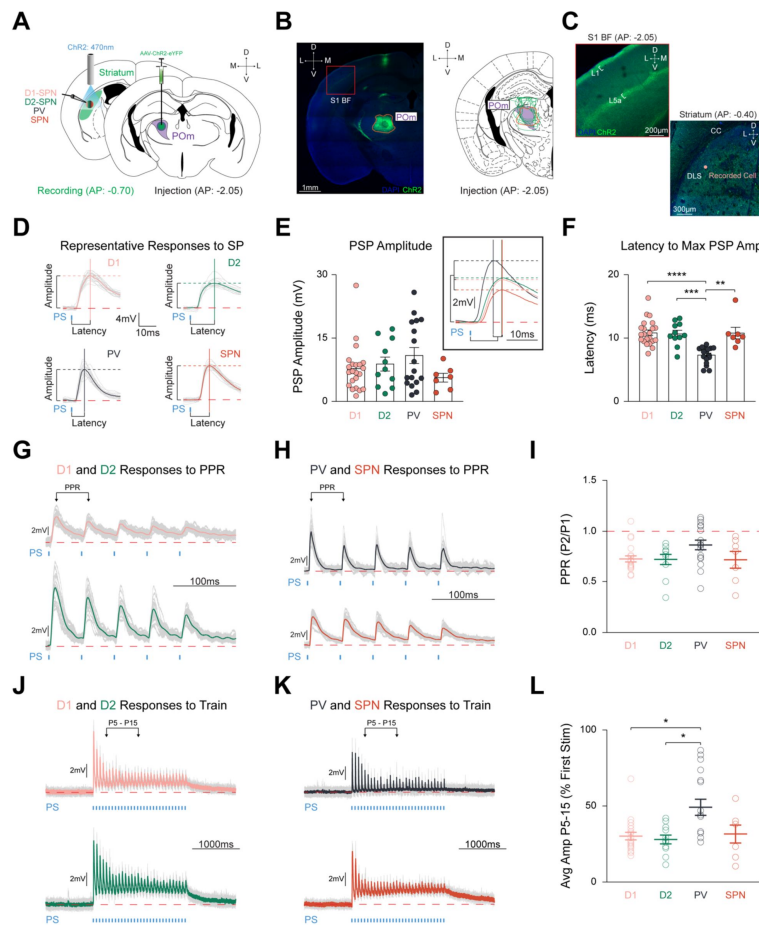


Figure 1.

POM Equally Innervates Striatal Cell Types with Faster Latency in PV Interneurons

(A) Schematic detailing pAAV-ChR2-EYFP injection unilaterally into POM (*Right*), and optogenetic stimulation of POM-striatal afferents whilst recording from identified and unidentified neurons via *ex vivo* slice of posterior DLS (AP range: -0.34 to -1.22 relative to Bregma; *Left*). See [Figure S1](#). Illumination (2.5ms pulses of 470nm light, ~0.6mW intensity) was delivered through the 40x objective.

(B) Representative injection site (orange) in POM (*Left*), and viral spread of all electrophysiology injections within highlighted POM (purple; *Right*). S1BF = S1 Barrel Field. Scale = 1mm.

(C) Red box inset from panel **(B)** highlighting stereotypical POM-cortical projection pattern to S1BF L1 and L5a. [25](#) [26](#) [49](#) *Right*: POM-striatal axons within posterior DLS. CC = corpus callosum. Scale = 200µm.

(D) Representative cell type-specific PSPs to SP stimulation. Colored lines = average PSP of 20 sweeps. Gray lines = 20 individual traces. Solid vertical and dashed horizontal lines = latency and amplitude, respectively. Red dashed line = 0mV. Blue tick = photostimulation (PS). Time scale = 10ms. Voltage scale = 4mV.

(E) Amplitudes evoked by each cell type were similar (D1-SPNs = 20 cells from 6 mice, D2-SPNs = 11 cells from 5 mice, PVs = 17 cells from 7 mice, unidentified SPNs = 7 cells from 4 mice). Inset shows grand average PSPs. Time scale = 10ms. Voltage scale = 2mV.

(F) Latency to maximum PSP amplitude is significantly quicker in PVs than all other cell types.

(G-H) Representative responses of **(G)** D1-SPN (*Top*) and D2-SPN (*Bottom*), and **(H)** PV (*Top*) and putative SPN (*Bottom*) to PPR stimulation. [109](#) PPR is defined as the ratio of PSP amplitude of pulse 2 over the ratio of PSP amplitude of pulse 1. PPR PS parameters = five 2.5ms pulses with 50ms interpulse intervals (20Hz). Time scale = 100ms. Voltage scale = 2mV. See [Figure S2I](#).

(J-K) Representative responses of **(J)** D1-SPN (*Top*) and D2-SPN (*Bottom*), and **(K)** PV (*Top*) and putative SPN (*Bottom*) to train stimulation. [111](#) Colored lines = average of 5 individual gray traces. Train PS parameters = thirty 2.5ms pulses with 64.2ms interpulse intervals (15Hz). Time scale = 1000ms. Voltage scale = 2mV.

(L) Relative PSP amplitude (average of pulses 5-15 compared to pulse 1) is significantly larger than both SPNs. Data are mean ± SEM. *p < 0.05, **p < 0.01, ***p < 0.001, ****p < 0.0001.

Short-Term Synaptic Dynamics Are Similar Between Striatal Cell Types, But Synaptic Depression Is Milder in PV Interneurons

The strength of synaptic inputs varies dramatically based on activation frequency and in a cell-type-specific manner with robust synaptic contacts generally exhibiting synaptic depression.^{89,90,107,108} To fully characterize the relative synaptic strength of POM-striatal inputs in a cell-type-specific manner, we assessed short-term plasticity by applying a PPR protocol of five pulses (**Figures 1G-H**, **S2A-B**).^{109,110} While all cell types exhibited robust synaptic depression overall, no PPR differences were observed (D1-SPNs: 0.73 ± 0.03 , $n = 20$, D2-SPNs: 0.72 ± 0.05 , $n = 11$, PVs: 0.86 ± 0.05 , $n = 17$, SPNs: 0.71 ± 0.08 , $n = 7$) ($F_{(4,51)} = 7.101$, $p = 0.0688$; **Figure 1I**).

To further characterize short-term synaptic dynamics, we applied a train protocol of thirty pulses at a frequency characteristic of POM-striatal activity (**Figure 1J-K**).^{18,111} Similar to PPR stimulation, train stimulation elicited overall synaptic depression in all cell types, but PV interneurons exhibited a milder synaptic depression relative to both SPN types that occurred predominantly between pulses 5-15 (**Figure 1L**, **S2**). PV interneurons (0.49 ± 0.05 , $n = 17$) showed significant differences compared to D1-SPNs (0.30 ± 0.03 , $n = 20$) and D2-SPNs (0.28 ± 0.03 , $n = 11$), but not with SPNs (0.31 ± 0.06) ($F_{(4,51)} = 10.99$, $p = 0.0118$, D1-SPN vs. PV $p = 0.0184$, D2-SPN vs. PV $p = 0.0481$, PV vs. SPN $p = 0.4940$; **Figure 1L**). Thus, POM-striatal projections provide SPNs and PV interneurons with unbiased and large amplitude synaptic inputs, characterized by milder synaptic depression in PVs, highlighting a potentially significant role in modulating striatal microcircuitry.^{89,90}

Mice Rapidly Learn to Discriminate Between Two Textures

While specific sensorimotor integrative and learning roles have been proposed and tested for several striatal inputs,^{69,72,87,89,90,104,112,113,114} the role of POM-striatal projections is still unknown. To monitor the activation of POM-striatal projections, we injected pAAV-axon-jGCaMP8s unilaterally into left POM and implanted a 400 μ m core cannula into left pDLS (**Figure 2A-B**). Violet light (405nm) and blue light (470nm) were constantly delivered to pDLS throughout the entire session to measure the isosbestic and POM-pDLS axonal calcium signals, respectively. Using a similar protocol from our previous publication,⁸⁹ water-restricted wild-type mice were trained on a whisker-based discrimination (Go/NoGo) paradigm. Mice received water for licking correctly (Hit) to the Go texture (P100 sandpaper). They received a white noise tone and a 12s time-out period for licking incorrectly (False Alarm; FA) to the NoGo texture (P1200 sandpaper; **Figure 2C**). If mice did not lick to the Go or NoGo texture, the texture retreated to its starting point, and trials were considered Miss and Correct Rejection (CR), respectively (**Figure 2D**). Additionally, pupil dynamics are a known metric of arousal,^{115,116} correlate well with POM activity,^{18,26} and exhibit outcome-dependent differences during the Go/NoGo paradigm.¹¹⁷ Therefore, synchronized orofacial video was captured during behavioral performance, and deep-learning^{118,119} pupillometry¹²⁰ was applied to assess pupil dynamics during task performance (**Figure S3B**) in which our results mirrored previously reported outcome-dependent differences.¹¹⁷

Mice in the fiber photometry (FP) cohort ($n = 5$) underwent three training phases (Shaping, Learning, and Expert; **Figure S3A**) that were segmented into five discrete behavioral time points (Shaping, Early Learning, Late Learning, Expert, and Reward; see Methods). During the learning phase, Hit rate increased, and FA rate decreased significantly, leading to markedly increased sensitivity (d') and slightly decreased bias (**Figure 2E**, **S3A**). Mice were considered Expert once they had reached ≥ 0.80 Hit Rate and ≤ 0.30 FA Rate for two consecutive sessions in lieu of a strict sensitivity (d') threshold; we found this definition more intuitive because d' is enhanced as

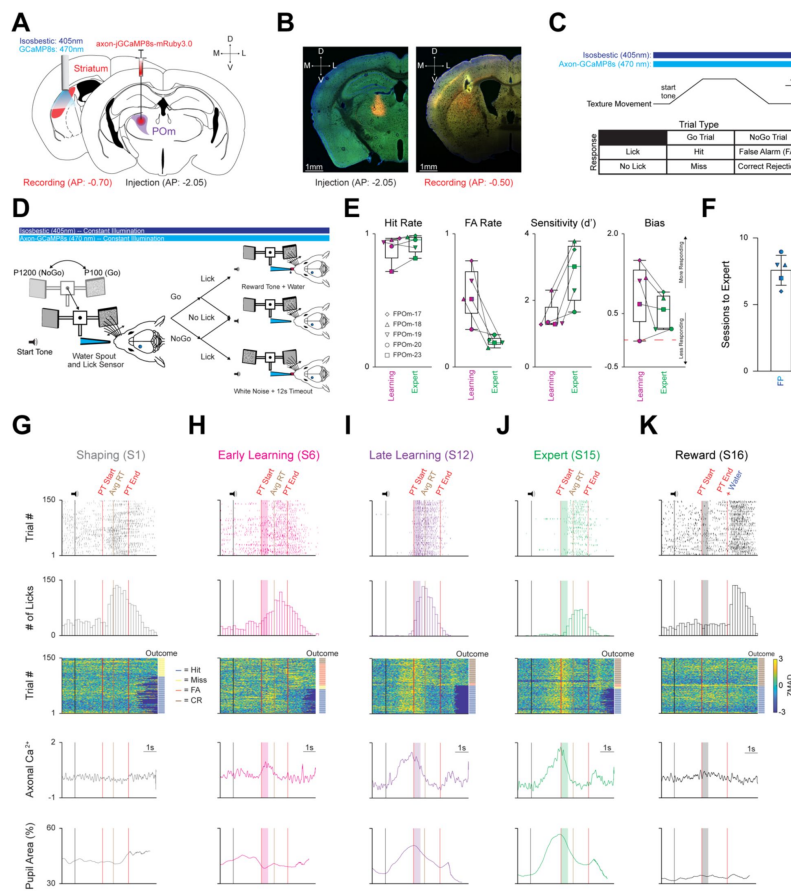


Figure 2.

Mice Rapidly Learn to Discriminate Between Two Textures, and All Three Activity Parameters Markedly Increase Across Learning

(A) Schematic detailing pAAV-hSynapsin1-axon-jGCaMP8s-P2A-mRuby3 injection unilaterally into POM (Right), and a 400µm cannula implanted in the left posterior DLS (Left).

(B) Representative injection site in POM (Left), and cannula placement in the posterior DLS along with ascending POM axons (Right). Scale = 1mm.

(C) Top: Stimulating timing and texture movement representation during a trial. Note = both LEDs (isobestic = 405nm; axon-jGCaMP8s = 470nm) were constantly on for every session. Bottom: Outcomes for each stimulus-response pair.

(D) Schematic representing texture movement and potential outcomes during a single trial of the Go/NoGo whisker discrimination paradigm.

(E) Changes in Hit Rate, FA Rate, Sensitivity (d') and Bias of the FP cohort ($n = 5$ mice) as they transition from the Learning to the Expert phase. Note that mice are classified as Expert when they achieve a Hit Rate ≥ 0.80 and a FA Rate ≤ 0.30 for two consecutive sessions. Red line = 0. See Figure S3.

(F) Average number of sessions required for expert discrimination of the FP cohort.

(G-K) Three activity parameters (licking, axonal calcium, and pupil activity) from a representative (G) Shaping (session 1), (H) Early Learning (first two sessions after Shaping), (I) Late Learning (last two sessions before Expert), (J) Expert, and (K) Reward sessions from the same mouse (FPom-18). Top: licking activity within a session (150 trials). Colored ticks = lick. Vertical black line = sound cue representing trial start as the texture moves towards the whisker field. Vertical red lines = start (texture arrival at endpoint in whisker field) and end (texture departure towards starting point) of the PT window (time where mice can respond by licking). Vertical brown line = average reaction time (RT; time of first lick that triggers an outcome) across all trials in each session. Colored boxes = 500ms grace period (licking does not trigger any outcomes). Note = no response line is present in the Reward session (K) as licking does not trigger any outcomes, and water was automatically delivered at PT end. Top Middle: Lick histogram. Middle: Heatmap sorted by trial outcome (to the Right of heatmap) highlighting axonal ZMAD calcium activity for each trial. Trial outcome is color coded (blue = Hit, yellow = Miss, orange = FA, brown = CR). Bottom Middle: Average axonal calcium activity of 150 trials for each session. Bottom: average pupil area (as a percentage) of 150 trials for each session. Data are mean $\pm$ SEM. Time scale = 1s.

Hit Rate and FA Rate approach their extremes (0 or 1) (**Figure S3A**). On average, it took this cohort 7.6 ± 0.51 sessions to become Experts (**Figure 2F**). Thus, the FP cohort rapidly learned to discriminate between the two textures as validated by behavioral responding parameters.

Calcium Activity Markedly Increases and Becomes Stereotyped Across Learning

To elucidate the contribution of POM-striatal projections during the Go/NoGo discrimination task, we measured POM axonal calcium activity along with licking and pupil activity at five discrete behavioral time points (Shaping, Early Learning, Late Learning, Expert, and Reward; **Figure 2G-K**). Photometry measurements revealed learning-related increases in POM axonal activity, starting before and peaking near texture presentation (**Figures 3A**, **S3A**) as measured by two parameters: calcium signal amplitude (ZMAD; median absolute deviation of the z-score) and area under the curve of the receiver-operator characteristic (auROC). First, the average maximum calcium amplitude significantly increased across learning ($F_{(1.818, 7.271)} = 39.24$, $p = 0.0001$, $n = 5$; **Figure 3B**). Further, the average maximum amplitude at Shaping (0.27 ± 0.04) was significantly smaller compared to Early Learning (1.04 ± 0.12 , $p = 0.0190$), Late Learning (1.58 ± 0.18 , $p = 0.0061$), and Expert (1.59 ± 0.17 , $p = 0.0052$). To test whether POM activity was reward-related, a single session was performed following the Expert phase during which the textures were removed, and water was automatically provided at the end of the presentation time (PT) window. We observed that calcium activity regressed to Shaping levels and was significantly smaller than Early ($p = 0.0306$), Late ($p = 0.0126$), and Expert ($p = 0.0098$).

Second, to quantify POM axonal activity within the striatum relative to learning, we employed signal detection theory, utilizing auROC values to compare the basal activity across the five behavioral time points as learning occurs.^{121,122} As with the maximum calcium amplitude, auROC values significantly increased across learning ($F_{(2.162, 8.650)} = 51.17$, $p < 0.0001$, $n = 5$; **Figure 3C**). Notably, auROC values at Shaping (0.54 ± 0.01) were significantly lower than Early Learning (0.70 ± 0.03 , $p = 0.0116$), Late Learning (0.83 ± 0.03 , $p = 0.0017$), and Expert (0.85 ± 0.02 , $p = 0.0007$), but not Reward (0.57 ± 0.01 , $p = 0.4111$). Further, significant auROC differences were also noted between the Early Learning vs. Expert ($p = 0.0383$), Late vs. Reward ($p = 0.0084$), and Expert vs. Reward ($p = 0.0042$). Finally, calcium-related events increased longitudinally and became stereotyped within a 4s target window (centered around texture presentation) compared to a non-task-related 1s control window (during the 1s pre-task interval prior to the sound cue and texture movement; **Figure S4A-D**). Thus, POM-striatal projections represent a learning-related signal that increases prior to and becomes stereotyped to texture presentation.

Licking and Pupil Dilation Also Markedly Increase Across Learning, But Only Pupil Is Correlated with Calcium Activity

Both licking and pupil dilation also exhibited marked increases across learning (**Figures 2G-K**, **S3A-B**). Licks occurring before texture presentation decreased dramatically as licking became stereotyped within the PT window (**Figure 3D**). We assessed whether licking and POM activity were correlated as both increase across learning (**Figure 3A, D**) by plotting the average calcium activity from the Early, Late, and Expert time points with the grand average reaction time (RT) set to 0 (**Figure 3E**). Comparison of average calcium activity pre- and post-RT revealed an overall significant reduction (0.79 ± 0.09 in pre-RT vs. 0.00 ± 0.03 in post-RT, $p = 0.0221$; **Figure 3F**), indicating no correlation between POM and licking activity, validating previous results.²⁶

Pupil dilation started immediately following the cue sound and peaked near texture presentation before slowly decreasing throughout the rest of the trial (**Figure 3G**). Similar to previous studies,^{18,26} we found that pupil and POM activity were tightly correlated but decoupled during the PT window with POM activity immediately regressing to baseline, while pupil activity remained elevated (**Figure 3A, G**). Pupil dilation lagged by ~ 250 ms on average (**Figure 3H**),

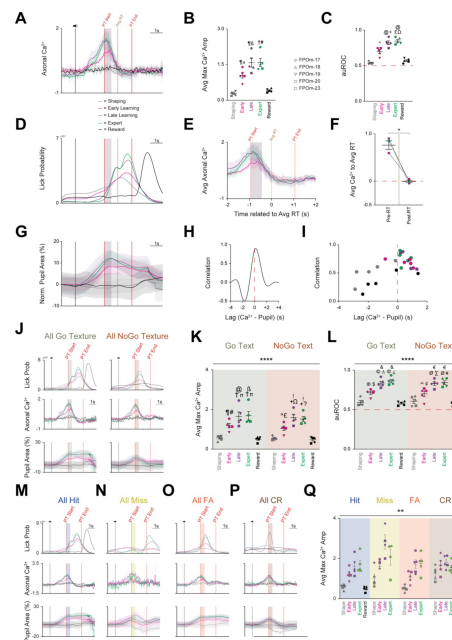


Figure 3.

All Three Activity Parameters Exhibit Marked Increases Across Learning, But Only Axonal Calcium Activity Remains Unchanged, Irrespective of Trial Type or Outcome Segmentation

- (A) Grand average axonal calcium activity at each behavioral time point. Data are mean $\pm$ SD.
- (B) Average of maximal axonal calcium amplitude markedly increases across learning before regressing to Shaping levels during the Reward session. $n p < 0.01$ Shaping vs. Early, $\beta p < 0.001$ Shaping vs. Late, $\# p < 0.001$ Shaping vs. Expert, $\P p < 0.05$ Early/Late vs. Reward, $\dagger p < 0.001$ Expert vs. Reward.
- (C) Average area under the curve of the receiver-operator characteristic (auROC) also markedly increases across learning before regressing to Shaping levels during the Reward session. $\wedge p < 0.05$, Shaping vs. Early; $+ p < 0.01$, Shaping vs. Late; $\pounds p < 0.001$, Shaping vs. Expert; $\Omega p < 0.05$, Early vs. Expert; $@ p < 0.01$, Late/Expert vs. Reward.
- (D) Grand average probability density function for licking-related activity at each time point.
- (E) Axonal calcium activity at Learning and Expert time points 2s pre and 2s post grand average RT. Data are mean $\pm$ SD.
- (F) Pre-RT axonal calcium activity is significantly larger than post-RT axonal calcium activity.
- (G) Grand average of normalized pupil area at each behavioral time point.
- (H) Representative cross-correlation of pupil area and axonal calcium activity plotted for each behavioral time point for each mouse.
- (I) Cross-correlation of pupil area and calcium activity plotted for each behavioral time point for each mouse.
- (J) Grand average of all licking (*Top*), calcium (*Middle*), and normalized pupil (*Bottom*) activity segmented by trial type: Go texture (*Left*) and NoGo texture (*Right*) presentation.
- (K) Average of maximal axonal calcium amplitude markedly increases across learning for both Go and NoGo texture presentation before regressing to Shaping levels during the Reward session. $\P p = 0.0016$ Go Shaping vs. Go Early, $n p < 0.0001$ Go Shaping vs. Go Late/Expert, $@ p = 0.0265$ Go Early vs. Go Late, $\beta p = 0.0154$ Go Early vs. Go Expert, $\# p = 0.0005$ Go Early vs. Go Reward, $\dagger p < 0.0001$ Go Late/Expert vs. Go Reward. $\wedge p = 0.0128$ NoGo Shape vs. NoGo Early, $+ p < 0.0001$, NoGo Shape vs. NoGo Late/Expert, $\pounds p = 0.0031$ NoGo Early vs. NoGo Reward, $\Omega p = 0.0085$ NoGo Early vs. NoGo Late, $p = 0.0295$ NoGo Early vs. NoGo Expert, $! p < 0.0001$, NoGo Late/Expert vs. NoGo Reward.
- (L) Average auROC markedly increases across learning for both Go and NoGo texture presentation before regressing to Shaping levels during the Reward session. $@ p = 0.0041$ Go Shaping vs. Go Early, $\textcircled{C} p < 0.0001$ Go Shaping vs. Go Late/Expert, $\Delta p = 0.0041$ Go Early vs. Go Late, $\zeta p = 0.0003$ Go Early vs. Go Expert, $\$ p = 0.0014$ Go Early vs. Go Reward, $\& p < 0.0001$ Go Late/Expert vs. Go Reward. $\infty p = 0.0462$ NoGo Shaping vs. NoGo Early, $\emptyset p < 0.0001$ NoGo Shaping vs. NoGo Late/Expert, $\Sigma p = 0.0017$ NoGo Early vs. NoGo Late, $\Upsilon p = 0.0005$ NoGo Early vs. NoGo Expert, $\textcircled{C} p = 0.0142$ NoGo Early vs. NoGo Reward, $\textcircled{E} p < 0.0001$ NoGo Late/Expert vs. NoGo Reward.
- (M-P) Grand average of licking (*Top*), calcium (*Middle*), and normalized pupil (*Bottom*) activity segmented by trial outcomes: (M) Hit, (N) Miss, (O) FA, and (P) CR.
- (Q) Average of maximal axonal calcium amplitude of each mouse markedly increases across learning for all trials outcomes before regressing to Shaping levels during the Reward session. Data are mean $\pm$ SEM unless noted otherwise. $* p < 0.05$, $** p < 0.01$, $*** p < 0.0001$. See also [Figure S4](#).

and the correlation became more consistent across learning and was restricted to before and at texture presentation (**Figure 3I**). Thus, despite all three activity parameters increasing across learning, POM activity is not related to motor activity and is only correlated with pre-PT pupil dilation.

Increased POM Axonal Calcium Activity Is Independent of Trial Type or Outcome

This whole-session analysis did not account for sensory (texture) or outcome differences. To assess whether POM axonal activity is sensory-related (texture-specific), sessions were segmented by the presented texture (Go or NoGo; **Figure S4E-F**) and by trial outcome (below). Notably, licking and pupil dynamics have discrete activation patterns based on the presented texture.¹¹⁷ Licking activity in the Go condition occurs squarely within the PT window and is indistinguishable from whole-session licking activity (**Figure 3D**), while licking peaks near the end of the grace period in the NoGo condition (**Figure 3J**). Similarly, pupil activity starts to increase at the same point in both conditions but deviates at the end of the grace period. Conversely, POM axonal activity remained strikingly consistent between the Go and NoGo conditions (**Figure 3J**). This observation was further validated by comparing the average maximum calcium amplitude of each behavioral time point between the presented texture conditions. The main effect of behavioral time points was significant ($F_{(4,32)} = 57.16$, $p < 0.0001$), but the main effect of presented texture was not ($F_{(1,8)} = 0.3797$, $p = 0.5549$; **Figure 3K**). Post-hoc comparisons found that the maximum calcium amplitude during Shaping (Go: 0.55 ± 0.06 ; NoGo: 0.53 ± 0.04) was significantly reduced compared to Early Learning (Go: 1.18 ± 0.11 , $p = 0.0016$; NoGo: 1.05 ± 0.11 , $p = 0.0128$), Late Learning (Go: 1.65 ± 0.23 , $p < 0.0001$; NoGo: 1.59 ± 0.16 , $p < 0.0001$), and Expert sessions (Go: 1.68 ± 0.21 , $p < 0.0001$; NoGo: 1.52 ± 0.12 , $p < 0.0001$), but not the Reward session. The same overall effects were noted for the auROC analysis (main effect of behavioral time point: $F_{(4,32)} = 66.43$, $p < 0.0001$; main effect of presented texture: $F_{(1,8)} = 0.01694$, $p = 0.8997$; **Figure 3L**). As with the maximum amplitude, post-hoc comparison highlighted that the auROC value during Shaping (Go: 0.59 ± 0.02 ; NoGo: 0.61 ± 0.03) was significantly smaller than Early Learning (Go: 0.72 ± 0.02 , $p = 0.0035$; NoGo: 0.70 ± 0.04 , $p = 0.0355$), Late Learning (Go: 0.84 ± 0.02 , $p < 0.0001$; NoGo: 0.83 ± 0.02 , $p < 0.0001$), and Expert sessions (Go: 0.87 ± 0.02 , $p < 0.0001$; NoGo: 0.85 ± 0.02 , $p < 0.0001$), but not the Reward session. Thus, despite divergent licking and pupil activity as a function of the presented texture, axonal calcium activity remained unchanged, indicating POM-striatal projections do not encode vibrissae-specific sensory information.

We next tested whether POM calcium activity correlated with trial outcome (e.g., Hit, Miss, False Alarm, Correct Rejection; **Figure 2C-D**). Upon segmentation by trial outcome (**Figure S4G-J**), licking and pupil dynamics exhibited discrete outcome-dependent activity patterns,¹¹⁷ but calcium activity again remained remarkably consistent (**Figure 3M-P**). These results were further validated by comparing the average maximum calcium amplitude of each behavioral time point between the outcome conditions: the main effects of behavioral time point ($F_{(1,273,5,090)} = 22.09$, $p = 0.0043$) and trial outcome ($F_{(1,153, 4,613)} = 10.80$, $p = 0.0231$; **Figure 3Q**) were significant. However, the only significant post-hoc comparisons were between Hit-Shaping and CR-Shaping ($p = 0.0305$) and FA-Shaping and CR-Shaping ($p = 0.0114$). Overall, POM-striatal projections do not appear to encode texture- or outcome-specific information, suggesting that POM may represent a behaviorally relevant arousal-related role.

Optogenetic Suppression of POM-Striatal Axons During Behavior Slows Learning Rate

Our findings indicate that POM-striatal inputs may represent an arousal signal. To investigate this possibility, wild-type mice were divided into two cohorts: a No Stim and a photoinactivation JAWS cohort. For both cohorts, a 200 μ m fiber optic cannula was unilaterally implanted over left pDLS. However, only the JAWS cohort received an injection of the inhibitory actuator JAWS (**Figure 4A-**

B into ipsilateral POM, verified by the stereotypical POM-cortical projection pattern (**Figure 4B**). Both cohorts were trained on the Go/NoGo task (**Figures S3**, **5**). However, once the Learning phase started, POM-striatal axons of the JAWS cohort were photoinactivated for ~50% of trials per session until reaching the Expert phase. Photoinactivation occurred via an amber (617nm) LED (~7mW) for 2s of constant illumination centered around the PT window start (**Figure 4C-D**) where the maximum calcium signals were detected (**Figure 3A**). Both the No Stim and JAWS cohorts achieved Expert status (**Figure S5A-B**) by increasing Hit Rate, decreasing FA Rate, and with licking becoming stereotyped across learning (**Figures 4E-H**, **S5A-B**). However, photoinactivation of POM-striatal axons during the Learning phase resulted in the JAWS cohort requiring significantly more sessions (12.50 ± 0.50 sessions, $n = 4$ mice) to attain Expert status compared to the other two cohorts with different experimental conditions (FP with constant illumination across the entire session: 7.60 ± 0.51 sessions, $n = 5$ mice; No Stim with no illumination: 7.40 ± 0.24 sessions, $n = 5$ mice) ($F_{(3,11)} = 8.542$, $p = 0.0041$, JAWS vs. FP $p = 0.0470$; JAWS vs. No Stim $p = 0.0202$; FP vs. No Stim $p > 0.9999$; **Figure 4I-J**). We assessed whether photoinactivation modified behavioral responding parameters between the Learning and Expert phases (**Figure 3K**). The only significant effect was a decrease in overall responding (Bias) during the Learning phase (Off: 0.56 ± 0.32 , On: 0.48 ± 0.32 , $p = 0.0351$). Due to this, we assessed if inhibition resulted in slower RTs and found that the average RT was significantly slower in the On condition (Off: 0.87 ± 0.07 , On: 0.93 ± 0.07 , $p = 0.0124$). Thus, the suppression of this behaviorally relevant arousal signal resulted in more learning sessions due to delayed RTs.

Discussion

POM Involvement in Sensorimotor Processing

POM has received considerable attention for its potential roles in sensory/nociceptive processing, pain signaling, and cortical plasticity mediation. Its widespread connectivity with sensory and motor cortical areas (S1, S2, and M1) along with its strong modulation with behavioral state and arousal, suggests a significant, yet still undetermined role in sensory-guided behaviors. VPM thalamus, by contrast, projects to much more delimited domains of S1, with axons coursing through striatum without forming synapses. POM projections to the dorsal striatum arise from branches of the main axons bound for the cortex and terminate in a region of the dorsal striatum that overlaps with corticostriatal inputs from S1 and M1, potentially placing POM projections in a powerful position to influence sensorimotor integration. We aimed to elucidate the functional innervation pattern of POM onto striatal cell types, and the role of POM-striatal projections on behavioral performance and learning. Slice electrophysiology revealed strong POM-mediated synaptic inputs to D1-SPNs, D2-SPNs, and PV interneurons, with shorter latency responses in PVs. *In vivo* photometry recordings showed increasing activation of POM-striatal axons across task learning, with axonal signals and pupil dilation in expert mice increasing prior to and becoming stereotyped to stimulus presentation, independent of stimulus type or trial outcome. Photoinactivation of POM-striatal axons resulted in prolonged RTs and delayed learning, with more training sessions required to achieve expert behavioral performance. We discuss below the complex feedforward and feedback circuitry that POM is part of, and potential functional differences in POM-striatal projections compared to POM-cortical projections in sensorimotor behavior. We propose that POM could provide an early salience- or arousal-related “priming” signal to the pDLS that interacts with subsequent corticostriatal signals, leading up to behavioral action selection.

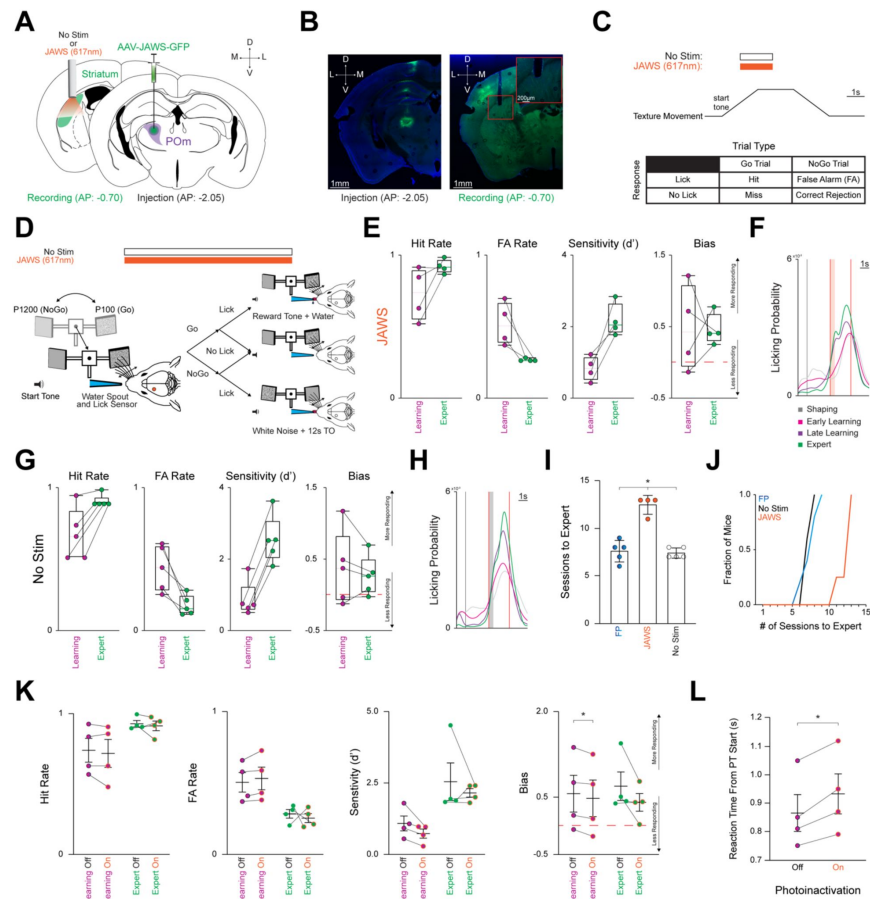


Figure 4.

Photoinactivation Increases Number of Sessions Required For Expert Discrimination

(A) Schematic detailing pAAV-hSyn-JAWS-KGC-GFP-ER2 (JAWS) injection unilaterally into POm (Right) and a 200µm cannula implanted in the left posterior DLS (Left). For the No Stim cohort, only the cannula was implanted in the left posterior DLS. Activation of the inhibitory JAWS opsin was performed constantly on for 1s before and after texture arrival in the whisker field. JAWS activation probability per trial = 0.50.

(B) Representative injection site in POm (Left), and the cannula placement in the posterior DLS along with ascending POm axons (Right). Scale = 1mm. Red inset shows ascending POm axons underneath the optic cannula. Inset scale = 200µm.

(C) *Top*: Stimulation timing (constant illumination for 2s, centered around texture arriving at its endpoint) and texture movement representation during a trial. Note that no light is presented for the No Stim cohort as no stimulation occurred. *Bottom*: Outcomes for each stimulus-response pair.

(D) Schematic representing texture movement and potential outcomes during a single trial.

(E) Changes in Hit Rate, FA Rate, Sensitivity (d'), and Bias of all JAWS cohort mice ($n = 4$) as they transition from the Learning to the Expert phase in box-and-whisker plots. Note that mice are classified as Expert when they achieve a Hit Rate ≥ 0.80 and a FA Rate ≤ 0.30 for two consecutive sessions. Red line = 0. See Figure S5.

(F) Probability density function for overall licking-related activity at each behavioral time point for the JAWS cohort. Vertical black line = sound cue representing trial start as the texture moves towards the whisker field. Vertical red lines = start (texture arrival at endpoint in whisker field) and end (texture departure towards starting point) of the PT window (time where mice can respond by licking). Colored boxes = 500ms grace period (licking does not trigger any outcomes).

(G-H) Same as in E, F for the No Stim cohort.

(I) JAWS cohort requires significantly more training sessions for expert discrimination compared to the FP and No Stim cohorts.

(J) Longitudinal representation of sessions required for expert discrimination.

(K) Comparison of Hit Rate, FA Rate, Sensitivity (d'), and Bias during the Learning and Expert phases.

(L) Average RT is slower during photoinactivation than non-photoinactivated trials. Data are mean $\pm$ SEM. * $p < 0.05$.

Does POM Provide a “Priming” Signal to Striatum?

A striking feature of our POM-striatal projection measures in behaving mice was the early task-related activation, with the strongest increases before and during texture presentation rather than during trial outcome, reward presentation, or licking. SPNs require an up-state transition from their relatively hyperpolarized resting potentials to reach action potential threshold.¹³¹ While this up-state transition is correlated with cortical oscillatory activity,^{87,132} the role of thalamostriatal projections has not been fully elucidated.¹³³ POM inputs likely arrive in striatum with shorter latency than cortical inputs following whisker stimulation,^{55,56,105,134} and therefore could be involved in initiating the up-state transition in the SPNs they innervate. We refer to this early POM-striatal input as a putative “priming” signal because it could serve to sensitize striatal neurons to respond to subsequent synaptic input. We found that POM provides large amplitude PSPs to D1- and D2-SPNs, of similar amplitude to previously measured M1 inputs and larger amplitude than S1 inputs.⁸⁹ This equally strong innervation of D1- and D2-SPN is consistent with previous work on the co-activation of SPN populations during natural movements.^{135,136,137} The early timing, together with the equal large-amplitude innervation of both direct and indirect pathways (D1- and D2-SPNs, and PV interneurons), suggest that POM projections are well-positioned to “prime” dorsal striatal circuitry for processing subsequent synaptic input.

Impact of PV-interneuron-mediated feedforward GABAergic signaling

POM innervation of PV interneurons was particularly prominent, providing equal amplitude PSPs with shorter latency than SPNs (**Fig. 1**). Striatal PV cells are thought to provide feedforward inhibition to surrounding neurons via short-latency GABAergic synaptic signals.⁶⁷ How could PV cells be involved in “priming”? First, PV cells do not exert an inhibitory effect, as defined by postsynaptic hyperpolarization, under all conditions. Stimulation of a presynaptic PV cell in synaptically coupled PV-SPN paired recordings causes GABA-mediated depolarization in the postsynaptic SPN when the SPN’s resting potentials was negative.¹⁷⁸ Indeed, recent work found that the chloride equilibrium potential is surprisingly positive in adult striatal SPNs, closer to -60 mV compared to the more typical -80 mV.¹³⁸ An approximately -60 mV equilibrium potential means that a GABA receptor-mediated chloride current produces depolarization instead of hyperpolarization when the postsynaptic neuron is at a down-state resting potential of approximately -80 mV.^{138,178,179} In Day et al. (2024), when GABA-mediated depolarization was paired with further glutamate-evoked depolarization, the probability of action potential firing was increased, suggesting that GABA-mediated depolarization was excitatory rather than shunting.¹³⁸ In this scenario, if SPNs are sitting at a relatively hyperpolarized (down state) resting potential when POM becomes active, the disynaptic GABAergic input provided by PV interneurons could actually have a depolarizing effect on SPNs and combine with the direct glutamate-mediated depolarization provided by POM. Thus, under certain conditions, PV interneurons could contribute to “priming” striatal circuitry by adding to the early depolarization of SPNs.

The situation is different when the postsynaptic neuron is already depolarized above -60 mV (as in an up state). In this scenario, the effect of GABAergic input is hyperpolarizing, and PV cell-mediated feedforward signals to SPNs would be inhibitory, in line with traditional views.^{63,65,66} In this case, instead of enhancing the initial depolarization of striatal neurons (as above), POM input would likely provide a temporally restricted depolarizing signal, followed by hyperpolarization. POM-mediated feedforward inhibition could, depending on the intricacies of local connectivity, act to prevent up-state transitions of specific SPN ensembles, effectively increasing the signal-to-noise of striatal population activity and therefore the recruitment of SPNs by subsequent cortical or thalamic inputs. Thus, it is possible that POM projections have dual

effects on striatal circuits depending on ongoing striatal activity. GABAergic signaling, mediated by PV or other types of striatal interneurons, is likely critical in mediating such state-dependent effects on postsynaptic striatal circuitry.

POM Involvement in Salience Networks and Behavioral State

Many extrinsic and intrinsic factors influence striatal function. For example, the striatum receives inputs from myriad subcortical regions including other thalamic nuclei^{73,99} and external globus pallidus.^{139,140} A crucial factor in POM signal timing is its inhibitory gating by two GABAergic inputs, ventral zona incerta (vZI)^{16,35,36,37,42} and anterior pretectal (APT).^{16,41,143} All three nuclei receive ascending spinal trigeminal (whisker-related sensory) inputs, but vZI and APT efficiently shunt incoming sensory information via feedforward inhibition onto POM neurons.^{16,39,44} This inhibitory gating is overcome by (1) arousal-related cholinergic suppression of presynaptic GABA release within POM⁴³ or (2) convergence of bottom-up and top-down signals within a specified temporal window.^{31,144} Both factors are likely at play during sensory-guided behavior. The involvement of cholinergic signaling is consistent with the POM-striatal “priming” concept if cholinergic signaling prompts POM before a behaviorally relevant period of sensation, or immediately prior to sensory information becoming perceptible.¹⁴⁵ In addition to cholinergic, cortical, and subcortical GABAergic afferents, POM also receives direct glutamatergic input from superior colliculus (SC) that enhances sensory responses.^{27,34} SC, a region implicated in attentional orienting of either somatosensory^{34,146} or visually-relevant^{147,148} stimuli, bidirectionally modulates POM, further ascribing a potential arousal-related functional role. It is yet to be resolved to what extent POM-striatal inputs are driven by ascending (feedforward) sensory information, descending cortical-POM feedback, or the convergence of both, but the interplay and integration of ascending and descending signals is likely to be an essential feature in POM activation and its involvement in behavioral salience.

It is difficult to differentiate movement- and arousal-related brain states. POM neuronal activity shows a close relationship with spontaneous (task-free) whisker movements, and pupil-indexed arousal in head-restrained mice,¹⁸ while in freely moving mice both VPM and POM activity correlate with head and whisker movements,¹⁷⁴ indicating that POM is generally coactive with exploratory head and whisker movements and behavioral arousal. During task performance, the situation may change with training and attentional effects. For example, Petty and Bruno (2024)¹⁷⁵ showed that POM activity correlates more closely with task demands than tactile or visual stimulus modality. Our data indicate that POM-striatal axonal signals are increased at trial start during anticipation of tactile stimulus delivery and through the sensory discrimination period, then decrease to baseline levels during licking and water reward collection (**Fig. 3**). Results of¹⁷⁵ together with ours suggest that POM is particularly active during the context of behaviorally relevant task performance. Thus, we think that pupil dilation indexes general movement and arousal, which POM correlates with during spontaneous behaviors, but that POM activity becomes more specific and more prominent during the movement, anticipation, and arousal associated with learned sensorimotor behavioral tasks.

Parallel Thalamostriatal Pathways, Additional Striatal Cell Types

Pf is the best studied thalamic nucleus projecting to the striatum. A major difference between POM and Pf striatal innervation is Pf innervation of cholinergic (ChAT; tonically active) interneurons.^{72,87,90,102,104,149,150} We did not determine whether POM innervates ChAT interneurons, but available evidence suggests it might only weakly. Retrograde tracing to map brain-wide inputs to striatal ChAT interneurons showed that Pf provides the majority of thalamic innervation of striatal ChAT neurons compared to other thalamic nuclei.¹⁴⁹ Many other thalamic nuclei, including POM, showed little or no labeling. However, it is possible that these thalamic nuclei, including POM, provide functional innervation of ChAT interneurons that is insufficiently assessed by anatomical methods alone.

Another difference between Pf and other thalamic nuclei innervation of striatum is the subcellular localization of their synaptic inputs. Pf synapses tend to localize to dendritic shafts of SPNs rather than dendritic spines^{151,152,153} whereas other thalamic nuclei form synapses onto SPN dendritic spines.¹⁷⁶ These differences could affect postsynaptic signal processing, such that Pf shaft inputs may induce a longer temporal window for integration via dendritic filtering,^{154,155} whereas POM spine inputs may cause a more restricted temporal window for signal integration. These properties could influence the behaviorally relevant time window of thalamostriatal and corticostriatal signal integration^{86,89,90}, for example, with POM requiring time-locked inputs onto the same or nearby spines for maximal postsynaptic effect.⁵⁶ The occurrence and relevance of such mechanisms in different thalamostriatal pathways will require further investigation. Additionally, it will be important to understand the innervation patterns of POM-striatal projections beyond the three cell types we have studied here, including other interneurons responsive to thalamic stimulation such as tyrosine-hydroxylase (THIN) and neurogliaform (NGF) interneurons.^{67,109} Lastly, the effects of dopamine modulation of thalamostriatal signaling directly⁶² and indirectly via cholinergic interneurons^{141,142} would be an important area of further study.

POM as a Link between Cortical and Striatal Signaling

POM axons bifurcate and contact neurons within pDLS,^{49,52,53,54,55,57,58} but the main terminations continue to cortex where they strongly innervate L1 and L5a of S1.^{48,49,54,55} POM input excites L5 pyramidal neurons (and L2/3 neurons) via L5 perisomatic synapses as well as distal dendritic synapses in L1.⁵⁰ Synaptic plasticity of POM-cortical inputs and increased excitation of L2/3 and L5 neurons has been implicated in sensory learning.^{22,23} Given the capacity of POM-cortical synapses for learning-related plasticity, it will be important to determine whether POM-striatal synapses also undergo learning-related plasticity and how it may be distinct from POM-cortical plasticity. Remarkably, POM axonal branches, even from the same neuron, display synaptic terminals with distinct structure and functional properties in M1 compared to S1¹⁷⁷; thus it is possible that the striatal synapses of POM neurons could undergo distinct changes with learning compared to their cortical counterparts.

How the divergent cortical and thalamic projections of POM neurons may be involved in coordinating thalamostriatal and corticostriatal signals is another important issue. Striatal neurons that receive POM input are likely also to receive cortical L5a input, because L5a neurons comprise the predominant S1-striatal projection.^{68,69,76} Thus a disynaptic POM-S1-pDLS loop exists in addition to the monosynaptic POM-pDLS projection. During behavior, striatal circuitry may be initially engaged via the monosynaptic POM projection followed, in tens of milliseconds, by the disynaptic loop that recruits L5a cortical pyramidal neurons to provide input to already depolarized (“primed”) striatal neurons. On the hundreds-of-milliseconds time scale of a behavioral trial, it is possible that these pathways are active multiple times, with L5b cortical feedback activating POM^{31,33} after the initial feedforward volley of activity, initiating POM-driven signaling again. It is possible that an individual SPN receives both POM and L5a input, with the POM input providing the initial “priming” depolarization, followed by further depolarizing input from a POM-driven L5a neuron. It is unclear whether thalamic “priming” of striatum is necessary for corticostriatal activation of SPNs. Based on the recurrence and convergence of these pathways, it is possible that POM acts to synchronize thalamo- and corticostriatal signals under certain behaviorally relevant conditions. This would be a unique role for POM compared to Pf, since Pf strongly innervates striatum but only weakly innervates cortex.

POM-striatal Recruitment during Learning: Unresolved Mechanisms

Behavioral state¹⁸ influences the efficiency of sensorimotor learning^{26,51}. Overall, we observed a learning-related increase in POM-striatal axonal activity (as measured by photometry) that correlated with pupil dilation in many but not all phases of behavioral performance, and a necessity of these projections for efficient behavioral performance and learning, supporting a role in task-related behavioral arousal. Photometry signals and pupil dilation in expert mice were tightly correlated between trial onset and texture presentation, but decoupled when photometry signals decreased prior to licking and reward delivery. Sorting trials by presented texture (stimulus) or trial outcome (response), photometry signals remained elevated, while both licking and pupil dilation exhibited stimulus- or response-specific changes. This suggests that POM-striatal projections do not encode specific sensory- or outcome-related information, but rather arousal or salience during anticipatory states of a learned behavior. The effects of photoinactivation of POM-striatal projections included delayed RTs on individual trials and more sessions required to achieve expert performance. These effects suggest that POM serves to enhance striatal signaling during increased behavioral states associated with performance of learned sensorimotor behavior. POM activation early during behavioral trials may set the stage for other important inputs, such as corticostriatal inputs, to influence action selection. The increased amplitude of POM-striatal signals with learning suggests an increased influence of POM on striatum, and thereby on goal-driven sensorimotor behaviors. POM likely becomes increasingly tuned to task-specific behavioral arousal and anticipation than to general movement-related arousal.

An important unresolved issue is how POM-striatal signals increase with learning. Inherent in our measures of POM-striatal axons is the limitation that photometry only captures a bulk axonal signal and lacks the ability to resolve individual axons. Therefore, we were unable to resolve whether the increased bulk signal that we observed was due to a uniform increase across all POM-striatal axons or a heterogeneous increase present only in specific axons. This is important because POM neurons exhibit heterogeneous responses to direct paralemniscal stimulation,¹⁰ peripheral stimulation,¹⁵ and the suppression of vZI activity,³⁹ and also show functionally distinct anterior and posterior subpopulations.^{2,49,54,130,156} Therefore, we might expect heterogeneous learning-related changes in activity within the POM neuronal population, a mechanistic question that would be important to address in future studies.

Although we placed the optical fiber above pDLS to specifically record from axons, rather than somas, it is possible that POM-cortical axons of passage contributed to the recorded signal. However, even if signal contamination were present, we think the measures would be largely similar, as most POM-cortical projections bifurcate within striatum rather than projecting exclusively to either striatum or cortex.^{49,55} Signal modification is more likely to occur at synapses via presynaptic or postsynaptic mechanisms than at the main axon.^{157,158,159} In future work it would be valuable to investigate the intriguing possibility that POM projections have discrete target-specific functions, such that POM-striatal signals may be involved in distinct aspects of sensorimotor behavior compared with POM-cortical signals.^{18,26,51}

In summary, we show that POM-striatal projections encode a behaviorally relevant arousal-related signal that increases with learning. POM-striatal projections may constitute a “priming” signal that combines with corticostriatal signals and contributes to inducing the up-state transition of SPN ensembles involved in action selection, by equally engaging D1- and D2-SPNs and PV interneurons. These findings suggest a previously unknown functional role of POM engagement of striatal microcircuitry. It will be important for future studies to investigate whether POM further innervates other striatal interneurons, to assess the timing between POM-striatal and corticostriatal inputs to SPNs, and to assess whether POM-striatal synapses undergo synaptic plasticity across learning.

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Author Contributions

A.J. Yonk and D.J. Margolis conceived and designed the research strategy. A.J. Yonk performed and analyzed electrophysiology experiments with assistance from M.A. Gradwell and A.J. George. A.J. Yonk performed and analyzed fiber photometry behavioral experiments with assistance from I. Linares-Garcia, L. Pasternak, S.E. Juliani, and A.J. George. A.J. Yonk performed and analyzed optogenetic behavioral experiments with assistance from I. Linares-Garcia and A.J. George. A.J. Yonk prepared the figures with input from I. Linares-Garcia, S.E. Juliani, M.A. Gradwell, A.J. George, and D.J. Margolis. A.J. Yonk wrote the paper, D.J. Margolis revised the paper after review, with all authors contributing to its editing. A.J. Yonk and D.J. Margolis acquired funding. D.J. Margolis supervised the study. All authors approved the final version of the manuscript.

Declaration of Interests

The authors declare no competing interests.

Methods

Data and Code Availability

Data are available upon reasonable request from the lead contact, David Margolis (david.margolis@rutgers.edu) and at github.com/margolislab.

Animals

All work involving animals including housing, surgery, behavioral experimentation, and euthanasia was approved by the Rutgers Institutional Animal Care and Use Committee (protocol #: 999900197). Mice were group housed in a reverse light cycle room (lights on from 20:00 to 08:00) with food and water available *ad libitum* with the exception of mice undergoing water restriction during behavioral experiments. All handling and experiments were conducted within the dark phase of this light cycle. Regardless of their experimental designation, all experimental animals underwent a unilateral AAV injection or a simultaneous AAV injection and cannula implant between -5 - 65 days (average: 52.37 days, range: -7 - 61 days). Briefly, male and female double transgenic mice were used for electrophysiology experiments. To identify specific neuronal populations during electrophysiology, Ai14 mice (cre-dependent tdTomato; The Jackson Laboratory, #007914) were crossed with either (1) D1-SPN cre (B6.FVB(Cg)-Tg(DrD1cre)EY262Gsat/Mmucd; MMRRC, #030989), (2) D2-SPN cre (B6.FVB(Cg)-Tg(Adora2a-cre)KG139Gsat/Mmucd; MMRRC, #036158), or (3) PV-cre (B6.129P2-Pvalb^{tm1(cre)Arbr}/J; The Jackson Laboratory, #017320) mice. This permitted red fluorescence in the specific cell population for visual identification via epifluorescent illumination. Animals designated for electrophysiology

were euthanized between 2 - 4.5 months. Both male and female wild-type mice (The Jackson Laboratory #000664) were used for fiber photometry and optogenetic experiments. To motivate behavioral performance, daily water intake was restricted to ~1.5mL per mouse per day. Body weight was carefully controlled and never permitted to drop below 80% of a calculated baseline value.¹⁶⁹ The number of male and female mice were as follows, by experiment type: 6 male, 4 female (electrophysiology); 3 male, 2 female (fiber photometry); 4 male, 5 female (optogenetics). Data were not analyzed for sex differences.

Electrophysiology and Adeno-Associated Viral (AAV) Injection

Male and female double transgenic mice designated for electrophysiology experiments underwent a unilateral injection of channelrhodopsin-2 (ChR2; pAAV-hSyn-hChR2(H134R)-EYFP; Addgene #26973)¹⁶⁰ targeting the left posterior medial (POM) thalamic nucleus. Briefly, mice were anesthetized using isoflurane (4% induction, 1-2% maintenance) and moved into a stereotaxic apparatus (Stoelting/Kopf Instruments) containing a feedback-controlled heating pad on the base (maintained between 35-37°C; FHC). Ophthalmic ointment (Akorn) was applied to the eyes to prevent them from drying out. Ethiqs XR (3.25 mg/kg; Fidelis Animal Health) and Bupivacaine (0.25%, 0.1mL, Fresenius Kabi) were injected subcutaneously into the right flank and scalp, respectively. After, the scalp was sterilized by three cycles of Betadine (Purdue Products) and 70% ethanol, a midline incision was made. The exposed skull was cleared of fascia and leveled by confirming that bregma and lambda coordinates were on the same dorsoventral plane. A craniotomy was drilled (coordinates with respect to bregma: anteroposterior (AP) = -2.05 mm, mediolateral (ML) = +1.35 mm, dorsoventral (DV) = -3.25 mm) and the micropipette containing ChR2 was slowly lowered to the appropriate depth and permitted to sit for 5 minutes. Following this, ~100 nL of ChR2 viral solution was injected over the course of ~15 minutes via the Nanoject III system (Drummond Scientific). After an additional delay of 12 minutes, the micropipette was slowly raised, the scalp was sutured and secured with tissue glue. Immediately following surgery, mice were transferred to a clean cage on top of a heating blanket until ambulation was observed. Mice were continually monitored for 72 hours post-surgery. After this observation period, mice were permitted to recover for at least three weeks before undergoing electrophysiological experiments, permitting ChR2 expression into POM axon terminals in the striatum.

Whole Cell Patch Clamp Recordings

Mice were briefly induced (via 3% isoflurane), deeply anesthetized with an intraperitoneal injection of ketamine-xylazine (300/30 mg/kg, respectively), and transcardially perfused with recovery artificial cerebrospinal fluid (ACSF), which contains the following (in mM): NMDG 103, KCl 2.5, NaH₂PO₄ 1.2, NaHCO₃ 30, HEPES 20, Glucose 25, HCl (1N) 101, MgSO₄ 10, Thiourea 2, Sodium Pyruvate 3, N-Acetyl-L-Cysteine 12, and CaCl₂ 0.5 (saturated with 95% O₂ and 5% CO₂).^{89,109} Following decapitation, the brain was rapidly extracted and submerged in recovery ACSF before being mounted onto a VT-1200S vibratome (Leica). The vibratome chamber was filled with oxygenated recovery ACSF, and 300µm slices were cut. Slices were immediately transferred to recovery ACSF that was heated to 35°C for ~5 minutes. After, slices were transferred to RT external ACSF which contained (in mM): NaCl 124, KCl 2.5, NaHCO₃ 26, NaH₂PO₄ 1.2, Glucose 10, Sodium Pyruvate 3, MgCl₂ 1, and CaCl₂ 2 (saturated with 95% O₂ and 5% CO₂), and slices were allowed to recover for at least ~45 minutes before recording. Once the hippocampus began to appear, vibratome sectioning was terminated, and the posterior tissue block containing the injection site was transferred into 10% neutral-buffered formalin for post-hoc confirmation.

Whole-cell patch clamp recordings were acquired from slices that were constantly superfused (2-4mL/min) with oxygenated external ACSF at ~34°C. Slices and cells were visualized by infrared differential interference contrast (IR-DIC) microscopy using a CCD camera (Hamamatsu) mounted onto a BX-51WI upright microscope (Olympus) fitted with a swinging objective holder containing

two switchable lenses: a 4X air lens and a 40X water-immersion lens. Patch pipettes (3–5 M Ω) were pulled from borosilicate glass micropipettes (2mm O.D., Warner Instruments) using a P-1000 horizontal puller (Sutter Instruments).

Current-clamp recordings were obtained from unidentified and identified striatal neurons in mice expressing tdTomato in either D1-SPNs, D2-SPNs, or PV cells within pDLS (–0.34 to –1.22mm relative to bregma), which is known to receive POM projections.^{55,56,105} The internal pipette solution for current-clamp experiments contained (in mM): K Methanesulfonate 130, KCl 10, HEPES 10, MgCl₂ 2, Na₂ATP 4, Na₂GTP 0.4, at pH 7.25 and 285–290 mOsm/L. Further, 2% biocytin was freshly dissolved in the internal solution on the recording day. ChR2 in the POM axon terminals was stimulated via illumination with a 2.5 ms, 470 nm LED light pulse (~0.6 mW measured after the objective; Thorlabs) delivered through the 40X objective lens. The illumination spot size had a diameter of ~550 μ m. After patching, the internal solution was permitted to dialyze for ~5 minutes. At this point, the resting membrane potential was recorded. All cells were held at -80 ± 2 mV to ensure equal driving forces when studying synaptic strength and short-term synaptic plasticity. Baseline voltages that drifted outside of this range were excluded from analysis. Patched cells were held for ~25–35 minutes while being run through a standardized set of protocols: (1) hyperpolarizing/depolarizing current steps, (2) single pulse (SP), (3) paired pulse ratio (PPR), and (4) train stimulation. Occasionally, unidentified cells within the same FOV were sequentially patched following the initial identified cell patch to control for injection site volume and location. After breaking in, the cell was allowed to recover for ~5 minutes before being subjected to hyperpolarizing and depolarizing current steps (–300 pA to +400 pA, 50 pA steps, 500 ms, 15 sweeps) for cell health and intrinsic parameter confirmation. For the SP protocol, a single 2.5 ms blue light pulse was presented once every 15s for 20 sweeps. For the PPR protocol, five 2.5 ms pulses, separated by 50 ms inter-pulse intervals (IPI), were presented once every 15s for 20 sweeps. For the train protocol, thirty 2.5 ms pulses, separated by 64.2 ms IPI (15 Hz), were presented once every 30s for 5 sweeps. While these protocols were being run, biocytin within the internal solution diffused into the cell for subsequent 3D morphological reconstructions. Data were acquired via a EPC10USB amplifier and digitized at 20 kHz in Patchmaster Next (HEKA). Liquid junction potential was not corrected in these traces.

Analysis of Patch Clamp Recordings

All data were analyzed offline using custom-written MATLAB and Python scripts. DAT files from Patchmaster Next (HEKA) were imported, organized, and saved as a mat variable. The mat variable data was imported into Python for post-processing using the electrophysiology feature extraction library eFEL created by the Blue Brain Project.¹⁶¹ All pertinent intrinsic value parameters were calculated at the +350pA current step. Key values pertaining to every action potential in a sweep were calculated including the action potential threshold value and index, the peak value and index, and the corresponding minimum afterhyperpolarization (AHP) value and index. Each action potential threshold value and index were captured using a derivative threshold method ($dV/dt \geq 10$ mV/s). Action potential properties were assessed for all spikes in a sweep and averaged together. The action potential peak was defined as the difference between the action potential threshold and its maximum positive peak. Half-height width (HHW) and rise time were both calculated via interpolation. HHW was measured at 50% of the action potential peak, while rise time was measured from 10–90% of the action potential peak. AHP amplitude was calculated as the difference between the action potential threshold value and the minimum AHP value. Maximum frequency

For all optogenetically-evoked postsynaptic potentials (PSPs), a baseline measure was calculated by averaging the first 10,000 sampling points for each individual sweep. This measured value was subtracted from every value in each individual sweep, setting the baseline equal to zero. The maximum PSP amplitude, relative to the zero baseline value, was calculated from the average of 20 sweeps during SP and PPR protocols, and 5 sweeps during Train protocols. The latency to maximum PSP amplitude was measured as the difference between photostimulation onset and the

maximum PSP index. For PPR and Train protocols, an exponential function was fitted to the decay component of each PSP, and their amplitudes were extracted after subtracting the decay of preceding PSPs.

Simultaneous AAV Injection and Optical Cannula Implantation Surgery

For male and female wild-type in the designated fiber photometry cohort, a 400 μ m core optical cannula (ferrule OD = 2.5mm, length = 2mm, nA = 0.50; Thorlabs #CFM15L02) was implanted directly above the left pDLS along with a ~100nL unilateral injection of axon-jGCaMP8s (pAAV-hSynapsin-axon-jGCaMP8s-P2A-mRuby3; Addgene #172921)¹⁶² into the ipsilateral POM. Male and female wild-type mice designated for optogenetic manipulation were split into two cohorts: (1) the No Stim or (2) the JAWS cohort. Both optogenetic cohorts were implanted with a 200 μ m core optical cannula (ferrule OD = 1.25mm, length = 2mm, nA = 0.50, CFMLC52L02, Thorlabs) above the left pDLS along with a ~100nL AAV injection into the ipsilateral POM. However, the No Stim cohort was injected with the excitatory optogenetic actuator, ChR2 (pAAV-hSyn-hChR2(H134R)-EYFP, Addgene #26973),¹⁶⁰ while the JAWS cohort was injected with the inhibitory optogenetic actuator, JAWS (pAAV-hSyn-JAWS-KGC-GFP-ER2, Addgene #65014).^{123,124,125,163} Further, an HHMI head plate¹⁶⁴ was fitted to each mouse using methods previously described.^{89,117,165,166,167} Briefly, mice were anesthetized with isoflurane (4% induction, 0.8 – 1.5% maintenance) and mounted within a stereotaxic frame (Stoelting/Kopf Instruments) with a feedback-controlled heating pad (FHC) maintaining the body temperature between 35 – 37°C. Following administration of an analgesic (Ethiqa XR, 3.25mg/kg; Fidelis Animal Health) to the left flank and a local anesthetic (0.25% Bupivacaine; Fresenius Kabi) under the scalp, the scalp was sterilized with a triple cycle of Betadine (Purdue Products) followed by 70% ethanol (Fisher). A midline incision was made, and a circular piece of scalp was removed to expose the skull. Both lateral muscles and the nuchal muscle were separated from the skull. The skull was cleaned by gently scraping away the periosteum. A dental etch bonding agent (iBond; Heraeus Kulzer) was applied to the clean skull and cured with blue light for 20s twice. A ring of dental composite (Charisma; Heraeus Kulzer) was applied to the outer edge of the skull and cured with blue light. After ensuring that bregma and lambda coordinates were equal in the dorsoventral plane, two craniotomies were made: one above the left posterior striatum (AP = -0.80mm, ML = +2.8mm, DV = 1.8mm) for cannula implantation and the other above left POM (AP = -2.05mm, ML = +1.35mm, DV = -3.25mm) for the corresponding AAV injection. The AAV injection was always performed first. The micropipette containing axon-jGCaMP8s (fiber photometry cohort), ChR2 (No Stim cohort), or JAWS (JAWS cohort) was slowly lowered to the appropriate depth and permitted to sit for 5 minutes. ~100nL of AAV solution was injected over the course of ~15 minutes via the Nanoject III system (Drummond Scientific). After an additional delay of 12 minutes, the micropipette was slowly raised. Once the optic cannula was secured in the stereotaxic frame and lowered to the appropriate DV coordinate, it was secured with dental composite (Tetric Evoflow; Heraeus Kulzer) that was cured with blue light. The HHMI headpost was delicately secured on the posterior area of the charisma ring with cyanoacrylate and dental composite. Finally, a single layer of dental composite was used to build and secure the rest of the headcap before being cured four times with blue light for 20s each. The scalp was closed around the headcap by using cyanoacrylate.

Immediately after the optical implant and viral injection surgery, mice were placed in a sterile, clean cage that was half-on/half-off a heating pad until ambulation was observed. Mice were diligently monitored for 72 hours post-surgery. After this monitoring period, mice were transferred to a clean cage on a ventilated rack for at least three weeks prior to handling and water restriction.

Handling and Water Restriction

After a three-week waiting period, mice were placed under citric acid water restriction¹⁶⁸ during which mice were handled twice daily for one week. The bitterness of citric acid naturally causes mice to reduce their water consumption and, consequently, their weight while still having access to water. Initially, mice were acclimated to handling as researchers placed their hands into the cage for increasing amounts of time (e.g., 5 minutes to 10 minutes to 15 minutes). After mice became comfortable, they were held for increasing amounts of time (e.g., 2 minutes to 5 minutes to 10 minutes). Additionally, mice were acclimated to head fixation by holding their HHMI headposts (e.g., 30 seconds to 1 minute to 2 minutes), and they were allowed to freely explore the behavioral tube for 5 minutes per handling session. Finally, mice were headfixed for increasing amounts of time in the behavioral setup (e.g., 5 minutes to 10 minutes to 15 minutes). Water was provided via transfer pipette to comfortably acclimate mice to head fixation. The head-fix apparatus contained a tube (length = 15cm; inner diameter = 4cm) that was affixed to a custom metal platform (length = 17cm; width = 12cm). The platform also contained HHMI mounting arms and holders for head fixation¹⁶⁴. During the final day of handling, mice were transitioned to full water restriction as this permits greater control over motivation level. During behavioral testing, daily water intake was limited to ~1.5mL per day to motivate performance on the Go/NoGo paradigm described below. The baseline body weight was measured daily during water restriction, and overall body weight was not permitted to drop below 80% of the baseline weight, consistent with levels of restriction used to motivate performance¹⁶⁹.

Fiber Photometry Setup

Fiber photometry data were collected using a RZ10x lock-in amplifier within the Synapse suite (Tucker-Davis Technologies). This amplifier and accompanying Synapse software was used to control a custom-built optical benchtop through drivers (LEDD1B, Thorlabs) to modulate LED signals. Briefly, this optical benchtop consisted of a self-contained system of four 30mm cage cubes with integrated filter mounts (CM1-DCH/M, Thorlabs). A 405nm LED (M405L4, Thorlabs) and a 470nm LED (M470L3, Thorlabs) were mounted onto the first 30mm cage cube. The 405nm LED was used to extract the calcium-independent isosbestic signal, and the 470nm LED was used to acquire calcium-dependent axonal GCaMP signals during the Go/NoGo paradigm. The 405nm isosbestic signal was modulated at 210Hz, and the 470nm GCaMP signal was modulated at 330Hz. A 425nm dichroic longpass filter (DMLP425R, Thorlabs) in the first cage cube reflected the 405nm excitation light and permitted the 470nm light to pass through. As the light entered the second cage cube, both excitation lights were reflected by a 495nm dichroic longpass filter (495DCLP, 67-079, Edmund Optics) into the third cage cube. A 460/545nm bandpass filter (69013xv2, Chroma) reflected both excitation wavelengths down to the subject via a low autofluorescence patch cable (MAF3L1, core = 400µm, NA = 0.50, length = 1m, Thorlabs). This cable was attached onto the implanted optical cannula (see above) by a ceramic mating sleeve (Thorlabs). Isosbestic and axon-jGCaMP8s emissions were collected via the optic cannula and passed through the 460/545nm bandpass filter (69013xv2, Chroma) into the fourth. Finally, the emission fluorescence passed through the detection pathway to reach the RZ10x photosensors for online observation.

Orofacial Video Recording

POm activity has been well correlated with whisking and pupil activity^{18,26}. To analyze these dynamics, an LED driver controlled an IR spotlight that illuminated the contralateral eye and mystacial pad, and facial recordings were captured through an autofocus USB webcam (NexiGo N660P) at ~20fps within the Synapse software. To limit light pollution from outside sources (e.g., LED illumination within the brain/eye), an IR filter was placed in front of the webcam. Also, a shortpass emission filter at 750 nm (Chroma #ET750sp-2p8) was placed between the third and fourth cage cubes to prevent recorded IR light from overloading the photosensors.

Synchronization of Task-Related Components

The Synapse software suite (Tucker-Davis Technologies) permitted the synchronous recording of both emitted (isosbestic and calcium) signals and video frames. Furthermore, the LabVIEW system-controlled paradigm-related components (e.g., texture movement, lick thresholds, trial type, trial outcome, etc.) and recorded the resulting behavioral parameters (e.g., licking activity, trial type, and trial outcome). To synchronize these two data streams for post-hoc analysis, TTL pulses relating to the texture arrival at its endpoint (which indicates the start of the presentation time window) from the LabVIEW system were captured within the Synapse software.

In Vivo Optogenetics

For the JAWS cohort, a high-powered 617nm LED (M617F2, Thorlabs) and current driver (LEDD1B, Thorlabs) were used for photoinactivation of POM axons in the striatum.^{123,124,125,163} LED stimulation was provided at a probability of 0.50 for every session from the start of the Learning phase until mice reached the Expert phase.²⁶ The light intensity was measured at ~7mW at the tip of the fiber. Stimulation intensity was kept consistent between mice and days by measuring the intensity with an optical power meter (PMD-100D, Thorlabs) prior to the first session every day. Light was delivered to the thalamic afferents in pDLS through an optical fiber patchcord (M95L01, fiber core diameter: 200µm, length: 1m, nA: 0.50) connected to a 200µm core optical cannula (described above) via a mating sleeve (Thorlabs). A small piece of black heat shrink tubing (Qualtek) was placed over the cannula during LED testing to prevent stray light from illuminating the presented texture during the task. The LabVIEW system controlled a Pulse Pal¹⁷⁰ that activated the LED current driver to provide constant illumination for two seconds, evenly split 1s before and 1s after texture presentation, corresponding to the increased calcium activity observed during fiber photometry recordings.

For the No Stim cohort, a high-powered 470nm LED (Prizmatix) and current driver (Prizmatix) were used for optogenetic activation. Note that no LED stimulation was provided during the Learning or Expert phase. Testing occurred during the first 4 sessions after the 5 initial Shaping sessions and the last 4 sessions after mice reached Expert status. Sessions consisted of 50 baseline trials followed by 10 alternating blocks of 10 OFF and 10 ON trials. The light intensity was measured at ~5mW at the tip of the fiber and was kept consistent between mice and days by measuring the intensity with an optical power meter (PMD-100D, Thorlabs) prior to the first session on testing days. Light was delivered to the thalamic afferents in pDLS through an optical fiber patchcord (M73L01, fiber core diameter: 200µm, length: 1m, nA: 0.50, Thorlabs) connected to a 200µm optical cannula (described above) via a mating sleeve (Thorlabs). A small piece of black heat shrink tubing (Qualtek) was placed over the cannula during LED testing to prevent stray light from illuminating the presented texture during the task. The LabVIEW system controlled a Pulse Pal¹⁷⁰ that activated the LED current driver to provide pulsed illumination for 2s at 15Hz (matching the electrophysiology train photostimulation paradigm), evenly split 1s before and 1s after texture presentation.

Go/NoGo Whisker-Based Discrimination Paradigm

Headfixed, water restricted mice were trained to perform a whisker-based discrimination paradigm as two textures were presented unilaterally to the right whiskers in a randomized order based on a custom-written LabVIEW code (National Instruments). This code used transistor-transistor logic (TTL) pulses to control all aspects of the paradigm including a water delivery spout connected to a piezo film sensor (MSP1006-ND; Measurement Specialties), and a motorized linear stage (T-LSM100A; Zaber) with a stepper motor (T-NM17A04; Zaber) containing two windmill arms holding two different sandpaper textures (Go texture = 100 grit sandpaper, P100; NoGo texture = 1200 grit sandpaper, P1200; 3M) as previously described.^{89,117,165,166,167} Mice were trained to discriminate between these two textures by licking the piezo film sensor spout.

Mechanical spout displacement resulted in transient voltage changes, and a lick was defined as voltage changes crossing either an upper or lower threshold once. After a lick was detected within the appropriate response window, the LabVIEW software immediately delivered the appropriate output: for Go trials, mice received a small water reward via a solenoid valve (0127; Buerkert) through the piezo spout, and, for NoGo trials, mice received a timeout period with co-occurring white noise. This paradigm occurred within a darkened room to minimize non-tactile related cues. Both textures and the headfix apparatus were cleaned with 70% ethanol between each mouse. If mice were not performing the task, sessions could be ended early. Water could be automatically delivered (AutoReward or AR) by the experimenter following 20 consecutive trials without a response. Finally, a session could be terminated if a mouse did not lick when water was present on the end of the spout after three ARs.

Trials began with a 1000ms pre-task interval followed by a brief cue tone (100ms, 2039Hz) that accompanied windmill texture movement towards the mice. Once the windmill texture had reached a predetermined distance within the whisking field, mice had to respond within a 2000ms presentation time (PT) window. For the first 500ms of the PT window, a grace period was present where responses did not trigger appropriate outcomes to reduce impulsivity. If mice licked in response to the Go texture, the trial was considered a Hit and resulted in a water reward accompanied by a correct tone (2793Hz). If mice licked in response to the NoGo texture, the trial was considered a False Alarm (FA) and resulted in punishment parameters including a time-out period (12000ms) and an accompanying white noise during the time-out period. Air puffs were eschewed as a punishment parameter to permit continuous pupil dynamic recording. If mice did not lick to the Go or the NoGo texture, nothing occurred, and the trial was considered a Miss or a Correct Rejection (CR), respectively. Immediately after a lick was recorded or the 2000ms PT window had elapsed, the windmill texture retreated to its original position where the current texture either remained or the other texture was rotated into position. Finally, trials were separated by a 2000ms intertrial interval. Behavioral performance was tracked across texture discrimination sessions by computing multiple behaviorally-related parameters including Hit Rate, FA Rate, Sensitivity (d') and Bias.¹⁷¹ For the fiber photometry cohort, the 405nm and 470nm LEDs provided constant illumination throughout all trials. For the JAWS cohort, LED photoinactivation occurred at a trial probability of 0.50 for every session from the start of the Learning until the end of the Expert phase.²⁶ For the No Stim cohort, LED stimulation did not occur during training (i.e., during either the Learning or Expert phases). It only occurred for 4 sessions after the initial 5 shaping sessions, and 4 sessions after mice reached expert status.

This whisker-based discrimination paradigm lasted up to 3 weeks. The FP cohort were only tested once per day to limit photobleaching. The JAWS and No Stim cohorts were tested twice daily. Each session consisted of 150 trials. All cohorts progressed through three behavioral phases (Shaping, Learning, and Expert) that were segmented into five analytical time points (Shaping, Early Learning, Late Learning, and Expert). The Shaping time point was the same for all mice. For the first three sessions, mice were acclimated to reliably trigger water delivery by licking the water spout. During these sessions, neither texture was presented to the whiskers. After, mice proceeded to texture discrimination training still under the Shaping phase. For the last two sessions, both textures were presented simultaneously with a Go texture probability of 0.90 and 0.75, respectively. Go and NoGo trials were interleaved in a pseudorandom order determined by the LabVIEW software. After the 0.75 probability session, mice progressed into the Learning phase. For all following sessions, the Go texture probability was set to 0.50 with a maximum of three consecutive presentations of the same texture. Discrete behavioral time points were determined as follows. The Early Learning time point was considered the first two sessions of the Learning phase. The Late Learning time point was considered the last two sessions of the Learning phase before achieving expert status. This occurred when mice had a Hit Rate ≥ 0.80 and a FA Rate ≤ 0.30 for two consecutive sessions. A strict sensitivity threshold was not used due to artificially increased sensitivity (discrimination) values as Hit Rate and/or FA Rate approach their extremes (e.g., see FPOM-18 sessions 8 to 15 in Figure 6). After discrimination training (i.e., achieving expert

status), mice in the fiber photometry cohort were subjected to a single Reward session to assess calcium, licking, and pupil activity in the absence of texture input and licking-related outcomes. During this session, the Go trial probability remained at 0.50, and the Zaber motor moved the windmill texture holder towards the whisker field, but the textures were rotated out of whisker range. Further, the upper and lower thresholds were set so that licking could not trigger outcomes before the end of the PT window. A water reward was automatically delivered at the end of the PT window during Go trials only. A whisker trim control session was performed in a subset of mice to confirm that mice used their whiskers to discriminate as previously observed.⁸⁹

Histology

For electrophysiology experiments, the tissue block containing the injection site was stored overnight in 10% neutral-buffered formalin at 4°C. After, it was transferred into 0.2% PBS Azide at 4°C until sectioning. The tissue block was mounted onto a stage and sectioned in 0.1M PBS at a thickness of 100µm using a VT-1000 vibratome (Leica). Slices were mounted onto microscope slides using DAPI Fluoromount-G (Southern Biotech #0100-20) and coverslipped before confocal imaging.

Following all behavioral experiments, mice were anesthetized with an intraperitoneal injection of Ketamine-Xylazine (120mg/kg Ketamine, 24 mg/kg Xylazine) and transcardially perfused with PBS followed by 10% neutral-buffered formalin. The brain was delicately extricated and stored in 10% neutral-buffered formalin overnight at 4°C. Tissue was mounted onto a stage and sectioned at 100µm using a VT-1000 vibratome (Leica). Sections were mounted and coverslipped using DAPI Fluoromount-G (Southern Biotech #0100-20). Confocal images were acquired using a LSM800 confocal laser scanning microscope (Zeiss) for injection site location verification, cannula placement, and viral expression in POM axons within pDLS and stereotypical POM-cortical projections in S1 L1 and L5a of all experimental mice.^{4,25,26,47,49} All data were acquired using the Zen software suite (Zeiss).

Quantification and Statistical Analysis

Behavioral Responding Analysis

Behavioral performance was tracked across texture discrimination sessions by computing multiple behaviorally-related parameters including Hit Rate, FA Rate, Sensitivity (d') and Bias.¹⁷¹ Hit Rate was calculated as follows: $[\text{Hit} / (\text{Hit} + \text{Miss})]$, where hit is the number of correct Go trials and Miss is the number of incorrect Go trials. The FA Rate formula is similar to the Hit Rate formula except it replaces Hit with FA and Miss with CR: $[\text{FA} / (\text{FA} + \text{CR})]$. Sensitivity illustrates the ability to discriminate between the signal (Go texture) and the noise (NoGo texture), and it is derived from signal detection theory.¹⁷¹ It is calculated as follows: $[\text{normalized inverse}-(\text{Hit Rate}) - \text{normalized inverse}(\text{FA Rate})]$. Finally, Bias illustrates the overall responding bias, independent of trial type. It is calculated as follows: $[0.5 * (\text{normalized inverse}(\text{Hit Rate}) + \text{normalized inverse}(\text{FA Rate}))]$.

Pupil Analysis

After all data were captured, the recorded orofacial videos were analyzed using DeepLabCut, a deep learning model for pose estimation, to estimate pupil dynamics as mice learned to discriminate between the two textures.^{118,119} Briefly, a model¹²⁰ was created with eight markers circumscribing the pupil, permitting the estimation of the approximate pupil area for each frame. All videos were cropped to a smaller dimension (230 x 275 pixels) that focused on each mouse's face. For each mouse, one video was selected at varying behavioral time points, and 30 frames were extracted and manually labeled with the eight markers. Each marker corresponded to a pupil location: top, top right, right, bottom right, bottom, bottom left, left, and top left. Additionally, another marker was placed on a static location (e.g., the water spout) to label frames when blinking occurred. Overall, the initial training dataset contained 150 labeled frames,

and the model was trained on this dataset with a ResNet50-based neural network for 250,000 iterations. After, the initial training videos were analyzed to assess the performance of the model from its marker estimations. 25 outlier frames from the initial training videos were extracted, manually corrected, and merged with the initial dataset. The model was trained again on this 275 labeled frame dataset with a ResNet50-based neural network for 250,000 iterations. After evaluating the network, the train error was calculated at 1.24 pixels, and the test error was measured at 1.25 pixels. Once the model was successfully trained, all behavioral videos of the five mice were analyzed. To estimate pupil area, a python library (scikit-image)¹⁷² was used to fit an ellipse on the estimations of the eight markers generated by DeepLabCut. This ellipse model was used to predict the values of the vertices and co-vertices on the ellipse, permitting the calculation of the major and minor axes. Thus, these measurements were used to approximate the area of the pupil for each frame. To detect blinking behavior, the maximum pupil size in non-blinking conditions was calculated and used as a threshold to identify abnormally high pupil predictions. As such, pupil areas greater than 450 pixels were removed via interpolation. The pupil area data were saved as a csv file for importing into MATLAB.

Fiber Photometry Signal Processing

Custom-written MATLAB scripts were used for post processing of the fluorescent signals. Photobleaching was corrected in both the isosbestic and GCaMP signals using detrended lines of best fit and subtracting the line from all values. After, the isosbestic and GCaMP median absolute deviation of z-score (ZMAD) signals were calculated before subtracting the GCaMP signal from the isosbestic to remove calcium-independent artifacts.

Alignment of Task-Related Components

A custom-written MATLAB script imported the lick-related and overall responding data, the processed ZMAD GCaMP signal, and the processed pupil csv for alignment. The overall responding data (containing trial elements such as time of PT window start which is the TTL flag within Synapse) was converted from UNIX timecode into seconds to match the Synapse (containing the pupil video and processed GCaMP signal) time. Pupil data was resampled to align with GCaMP signal using rational fraction approximation. Furthermore, the length of each trial, as determined by the overall responding data, was recorded and used to capture the GCaMP and pupil data within each trial window. Licks were identified by setting upper and lower thresholds, and detecting when either was crossed. At this point, all parameters (e.g., GCaMP, pupil, and licking activity) were captured and aligned within the time window of each trial. These parameters could now be segmented by trial type (e.g., Go vs. NoGo texture) and by trial outcome (e.g., Hit, Miss, FA, CR) for more advanced analysis.

Calcium Analysis

auROC is an analysis commonly applied to calcium imaging data to characterize the stereotypy of neuronal responses.^{121,122} Briefly, a baseline window is set within a non-task-related component of the overall calcium signal and compared to the rest of the signal via a sliding window. The maximal value of each signal is captured at each behavioral time point and averaged across the cohort. auROC values equal to 0.50 indicate no differences between the baseline signal and the task-related signal.

To assess longitudinal changes in calcium activity, two windows were established: a Control Window located from trial start to the sound cue indicating trial start, and a Target Window located 2s before and after PT start (overall = 4s). For each trial, all calcium peaks were measured, and only peaks $\geq$ 90th percentile with a minimum peak prominence of 2 were selected. Finally, the maximum values within the Control and Target Windows (if present) were captured and averaged for every session.¹⁷³

Statistical Analyses

All data are reported as mean $\pm$ SEM unless otherwise noted. Statistical analyses were performed in GraphPad Prism (USA). All data were tested using the Shapiro-Wilk normality test. The means of different data distributions were analyzed and compared using two-tailed Student's t-test (**Figures S10, S1R** [↗](#), 2E FA Rate, 2E Bias, 3F, 4E Hit Rate, 4E Sensitivity, 4E Bias, 4G FA Rate, 4G Bias, 4K Learning/Expert Hit Rate, 4K Learning/Expert FA Rate, 4K Learning/Expert Sensitivity, 4K Learning/Expert Bias, 4L, S5C Hit Rate, S5C FA Rate, S5C Sensitivity, S5C Bias), Wilcoxon signed rank test (**Figures S1N, S1Q** [↗](#), 2E Hit Rate, 2E Sensitivity, 4E FA Rate, 4G Hit Rate, 4G Sensitivity), ordinary one-way ANOVA with Tukey's multiple comparison correction (**Figures S1G, S1L** [↗](#)), Kruskal-Wallis with Dunn's multiple comparison correction (**Figures 1E** [↗](#), 1F, 1I, 1L, S1H, S1I, S1J, S1K, 4I), linear regression (**Figure S1E** [↗](#)), Repeated measures one-way ANOVA with Tukey's multiple comparison correction (**Figures 3B** [↗](#), 3C), Repeated measures mixed-effects analysis with Tukey's multiple comparison correction (**Figures S4D Top** [↗](#), **S4D Bottom** [↗](#)), Repeated measures two-way ANOVA with Sidak's multiple comparison correction (**Figures 3K, 3L** [↗](#)), Repeated measures mixed-effects analysis with Sidak's multiple comparison correction (**Figure 3Q** [↗](#)). For all statistical tests, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$.

Supplemental Data

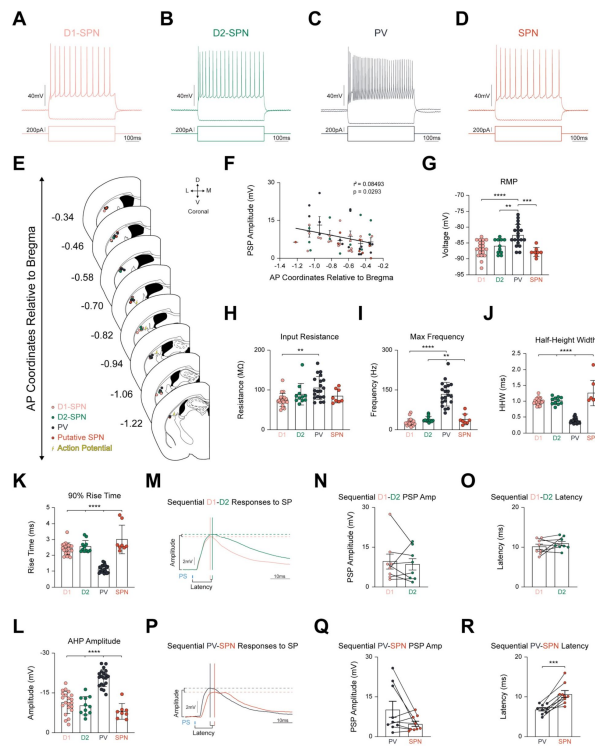


Figure S1.

Recorded Cell Location, Intrinsic Electrophysiological Parameters, and Sequentially Patched PSP Amplitude and Latency. Related to Figure 1

(A-D) Representative responses of (A) D1-SPN (salmon), (B) D2-SPN (green), (C) PV interneuron (dark gray), and (D) unidentified SPN (orange) to hyperpolarizing and depolarizing current injections. Vertical voltage scale = 40mV. Horizontal current scale = 100ms. Vertical current scale = 50pA.

(E) Recording location schematic of all recorded cells including cells excluded due to action potentials (lightning bolt). Note = all recordings took place within posterior DLS as it is the only striatal region innervated by POM axonal projections [S1, S2].

(F) Small but significant correlation ($r^2 = 0.08493$; $p = 0.0293$) between PSP amplitude and increasing distance from the injection site (AP = -2.05) excluding cells with action potentials.

(G-L) Parameters are consistent with literature concerning both SPNs and PV interneurons: (G) Mean resting membrane potential (RMP; D1-SPN: -86.80 ± 0.58 mV; D2-SPN: -86.00 ± 0.69 mV; PV: -82.55 ± 0.76 mV; SPN: -87.88 ± 0.52 mV) ($F_{(4,56)} = 11.24$, $p < 0.0001$, D1 vs. PV $p < 0.0001$, D2 vs. PV $p = 0.0079$, SPN vs. PV $p = 0.0001$), (H) Mean input resistance (R_{in} ; D1: 74.97 ± 3.60 M Ω , D2: 89.25 ± 8.25 M Ω , PV: 105.00 ± 6.51 M Ω , SPN: 84.58 ± 6.16 M Ω) ($F_{(4,56)} = 14.27$, $p = 0.0026$, D1 vs. PV $p = 0.0011$, I) Mean maximum frequency (D1: 26.34 ± 2.95 Hz, D2: 38.38 ± 3.46 Hz, PV: 134.00 ± 10.07 Hz, SPN: 38.00 ± 7.25 Hz) ($F_{(4,56)} = 42.64$, $p < 0.0001$, D1 vs. PV $p < 0.0001$, D2 vs. PV $p = 0.0032$, SPN vs. PV $p = 0.0027$), (J) Mean half-height width (HHW; D1: 0.99 ± 0.02 ms, D2: 1.01 ± 0.04 ms, PV: 0.38 ± 0.02 ms, SPN: 1.26 ± 0.14 ms) ($F_{(4,56)} = 42.06$, $p < 0.0001$, D1 vs. PV $p < 0.0001$, D2 vs. PV $p < 0.0001$, SPN vs. PV $p < 0.0001$), (K) Mean 90% rise time (D1: 2.42 ± 0.08 ms, D2: 2.56 ± 0.12 ms, PV: 1.14 ± 0.05 ms, SPN: 3.00 ± 0.32 ms) ($F_{(4,56)} = 40.52$, $p < 0.0001$, D1 vs. PV $p < 0.0001$, D2 vs. PV $p < 0.0001$, SPN vs. PV $p < 0.0001$), and (L) Mean afterhyperpolarization (AHP) amplitude (D1: -11.02 ± 0.91 mV, D2: -9.79 ± 1.05 mV, PV: -20.39 ± 0.73 mV, SPN: -7.64 ± 1.05 mV) ($F_{(4,56)} = 38.28$, $p < 0.0001$, D1 vs. PV $p < 0.0001$, D2 vs. PV $p < 0.0001$, SPN vs. PV $p < 0.0001$).

(M-O) Identified and unidentified neurons were patched sequentially and tested using the same protocols (SP, PPR, and Train) to control for injection site variability. (M) Representative PSP responses of a sequentially patched D1- and D2-SPN pairing ($n = 8$ D1-D2-SPN sequential recordings from 5 mice). Time scale = 10ms. Voltage scale = 2mV. (N) No significant differences in PSP amplitude between sequentially patched D1-SPN and D2-SPN pairs. (O) No significant differences in latency between sequentially patched D1-SPN and D2-SPN pairs.

(P) Same as in M for recording acquired from sequentially patched PV and SPN pairings ($n = 9$ PV-SPN sequential recordings from 7 mice). Time scale = 10ms. Voltage scale = 2mV.

(Q) No significant difference in PSP amplitude between sequentially patched PV and SPN pairs.

(R) Significant differences in latency between sequentially patched PV and SPN pairs (6.85 ± 0.42 for PV vs. 10.75 ± 0.87 for SPN, $p = 0.0006$, $n = 9$ pairs). Data are mean $\pm$ SEM. ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$.

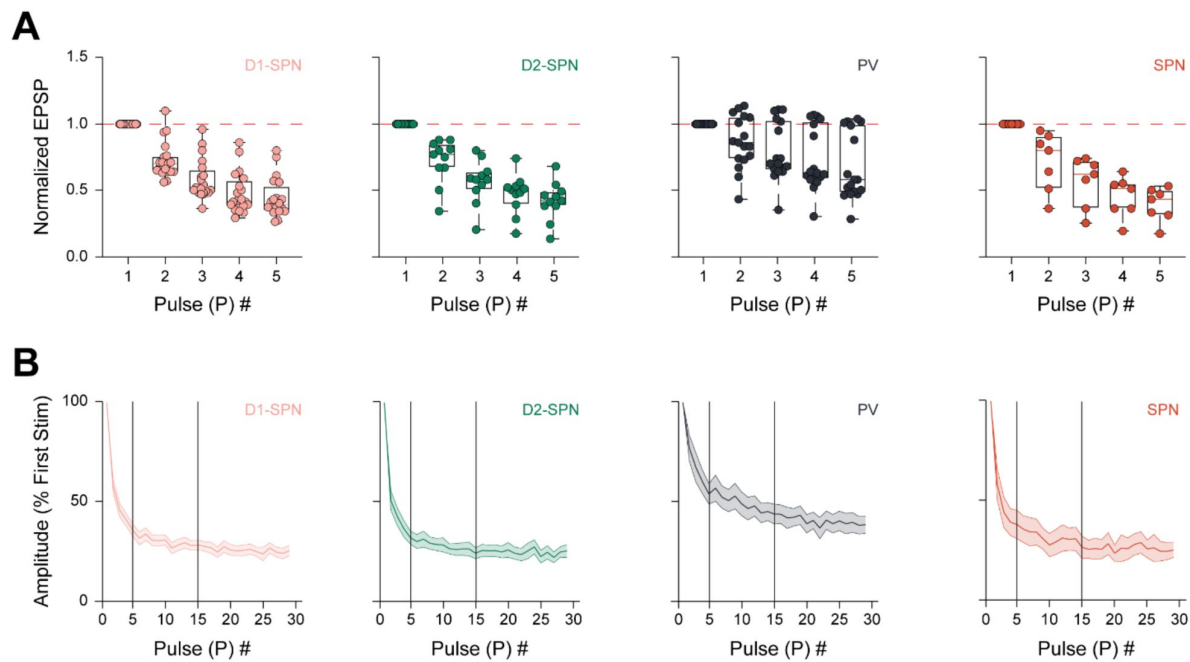


Figure S2.

Representative Responses of Identified Striatal Cells to PPR and Train Stimulation. Related to Figure 1.

(A) Box-and-whisker plot of the normalization of the second PSP to the first PSP of all recorded D1-SPNs (salmon), D2-SPNs (green), PV interneurons (dark gray), and unidentified SPNs (orange).

(B) Summary plots of all PSP amplitudes normalized to the first pulse for all recorded D1-SPNs, D2-SPNs, PV interneurons, and unidentified SPNs. Vertical black lines indicate the averaging within from pulse 5 to pulse 15. Data are mean $\pm$ SEM.

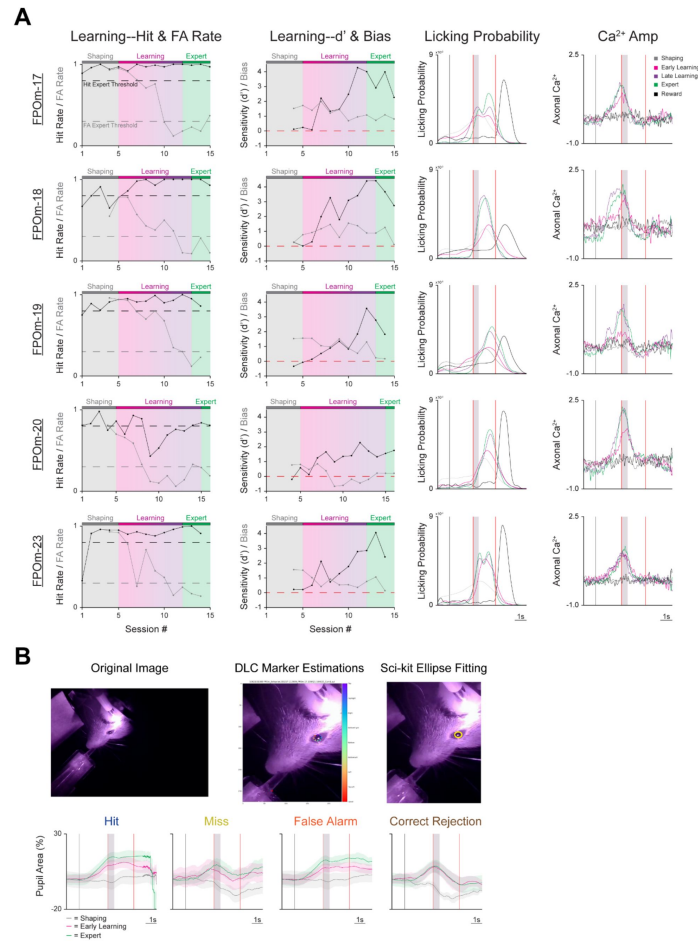


Figure S3.

Individual Longitudinal Learning-Related Changes in Behavioral Parameters, Licking Activity, and Calcium Activity, and Methodology of Measuring Pupil Dynamics. Related to Figure 2.

(A) Longitudinal learning-related changes as mice progress through each time point (Shaping, Early Learning, Late Learning, and Expert). The Shaping time point permitted mice to learn to lick the spout for a water reward during the first three sessions without the presence of the textures. During the last two Shaping sessions, the Go and NoGo textures were presented simultaneously with a Go texture probability of 0.90 for the first session and 0.75 for the second session. For the first Learning session, and all subsequent sessions, the Go texture probability was set to 0.50. Mice were considered Learning at this time point. The Early Learning time point was considered the first two sessions after Shaping, while the Late Learning was considered the last two sessions before achieving Expert status. Mice were considered Expert when they had a Hit Rate ≥ 0.80 and a FA Rate ≤ 0.30 for two consecutive sessions. After discrimination training was completed, mice were subjected to a Reward session. During the Reward session, the lick thresholds were unobtainable, the textures were oriented so they could not be contacted by the whiskers, and water was automatically delivered at the end of the PT window (second vertical red line). *Left*: Hit Rate (black) and FA Rate (gray) across learning. *Left Middle*: Sensitivity (d' ; black) and Bias (gray) across learning. Red dashed line indicates 0. Note that a strict d' threshold was not used due to artificially increased d' values as Hit Rate and/or FA Rate approached their extremes (0 or 1) as in FPOm-18 sessions 8 to 15. *Right Middle*: Probability density function for licking activity at each time point. *Right*: Average axonal ZMAD calcium activity at each time point. Scale bar = 1s.

(B) To measure pupil dynamics during the Go/NoGo discrimination task, orofacial video was synchronously recorded with calcium activity. This video was cropped in DeepLabCut [S3, S4], and nine markers were manually placed within the video: eight circumscribing the pupil (top, top right, right, bottom right, bottom, bottom left, left, and top left) and one on the spout. Once the model was trained, an ellipse was fitted through the eight pupil markers in sci-kit [S5]. Pupil values were converted from pixels to percentage, and the baseline was normalized to 0. **(B)** Notably, pupil-related dynamics (e.g. Shaping, Early Learning and Expert) during the trial outcomes (e.g. Hit, Miss, False Alarm, and Correct Rejection) are similar to previously observed dynamics [S6] despite using a deep learning-based methodology [S7]. Data are mean $\pm$ SEM. Scale bar = 1s.

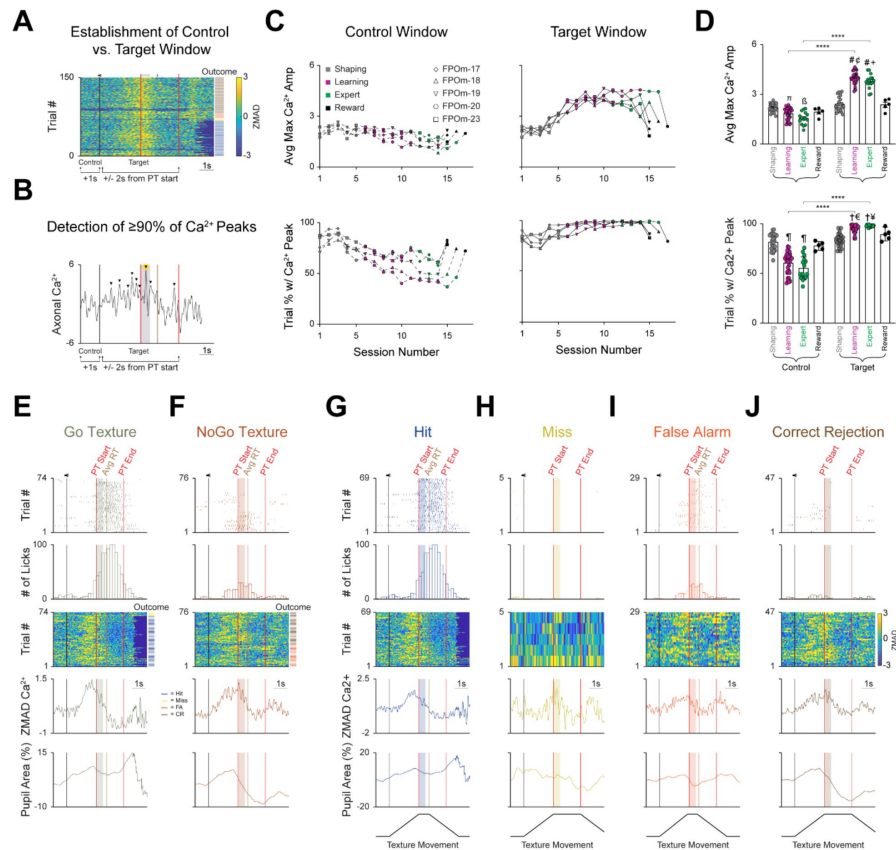


Figure S4.

Establishment of Control and Target Windows, and Representative Example of All Three Activity Parameters Segmented by Trial Type and Outcome. Related to Figure 3

(A) Establishment of the Control (1 second following trial start, encompassing the pre-trial intertrial interval) and Target (2 seconds before and after PT window start) windows. Task-related events are directly above the heatmap. Trial outcomes are color coded (blue = Hit, yellow = Miss, orange = False Alarm (FA), brown = Correct Rejection (CR)).

(B) Representative axonal calcium activity during a single trial (outcome = hit). Black arrowheads above positive deflections indicate a calcium peak that was greater than or equal to the 90th percentile of all calcium peaks. The yellow circle encompassing the largest calcium peak was selected as the maximal calcium amplitude.

(C) Longitudinal representation of the average maximum calcium amplitude (*Top*) and the average trial % with a detectable calcium peak (*Bottom*) within the Control (*Left*) and Target (*Right*) windows.

(D) Average maximum calcium amplitude (*Top*, $n = 0.0005$, Control Shaping vs. Control Learning; β $p < 0.0001$, Control Shaping vs. Control Expert; $\#$ $p < 0.0001$, Target Shaping vs. Target Learning/Expert; ζ $p = 0.0073$, Target Learning vs. Target Reward; $+$ $p = 0.0091$ Target Expert vs. Target Reward; Mixed-Effects Analysis with Tukey's correction for multiple comparisons). Average trial % with a detectable calcium peak (*Bottom*, $\P$ $p < 0.0001$ Control Shaping vs. Control Learning/Expert; $\dagger$ $p < 0.0001$, Target Shaping vs. Target Learning/Expert; ϵ $p = 0.0073$, Target Learning vs. Target Reward; Υ $p = 0.0091$, Target Expert vs. Target Reward; Mixed-Effects Analysis with Tukey's correction for multiple comparisons).

(E-F) Three activity parameters (licking, axonal calcium, and pupil) from a representative session segmented by trial type: (E) Go texture or (F) NoGo texture presentation. *Top*: Licking activity. Licks are denoted as colored tick marks. The vertical black line represents a sound cue indicating trial start as the presented texture moves towards the whisker field. The vertical red lines denote the start (texture arrival at its endpoint in the whisker field) and end of the PT window. The vertical brown line denotes average RT. The colored boxes denote a 500ms grace period wherein the mouse can lick freely without triggering any outcomes. *Top Middle*: Lick histogram. *Middle*: Heatmap sorted by trial outcome (*Right* of heatmap) highlighting axonal calcium activity for each trial. Trial outcome is color coded (Blue = Hit, yellow = Miss, Orange = FA, Brown = CR). *Bottom Middle*: Average axonal calcium activity. *Bottom*: Average pupil area (as a normalized percentage).

(G-J) The representative session from E-F was further segmented by trial outcome: (G) Hit, (H) Miss, (I) FA, and (J) CR. Underneath the *Bottom* panel, texture movement that is dependent on trial outcome is illustrated. Data are mean $\pm$ SEM.

*** $p < 0.0001$.

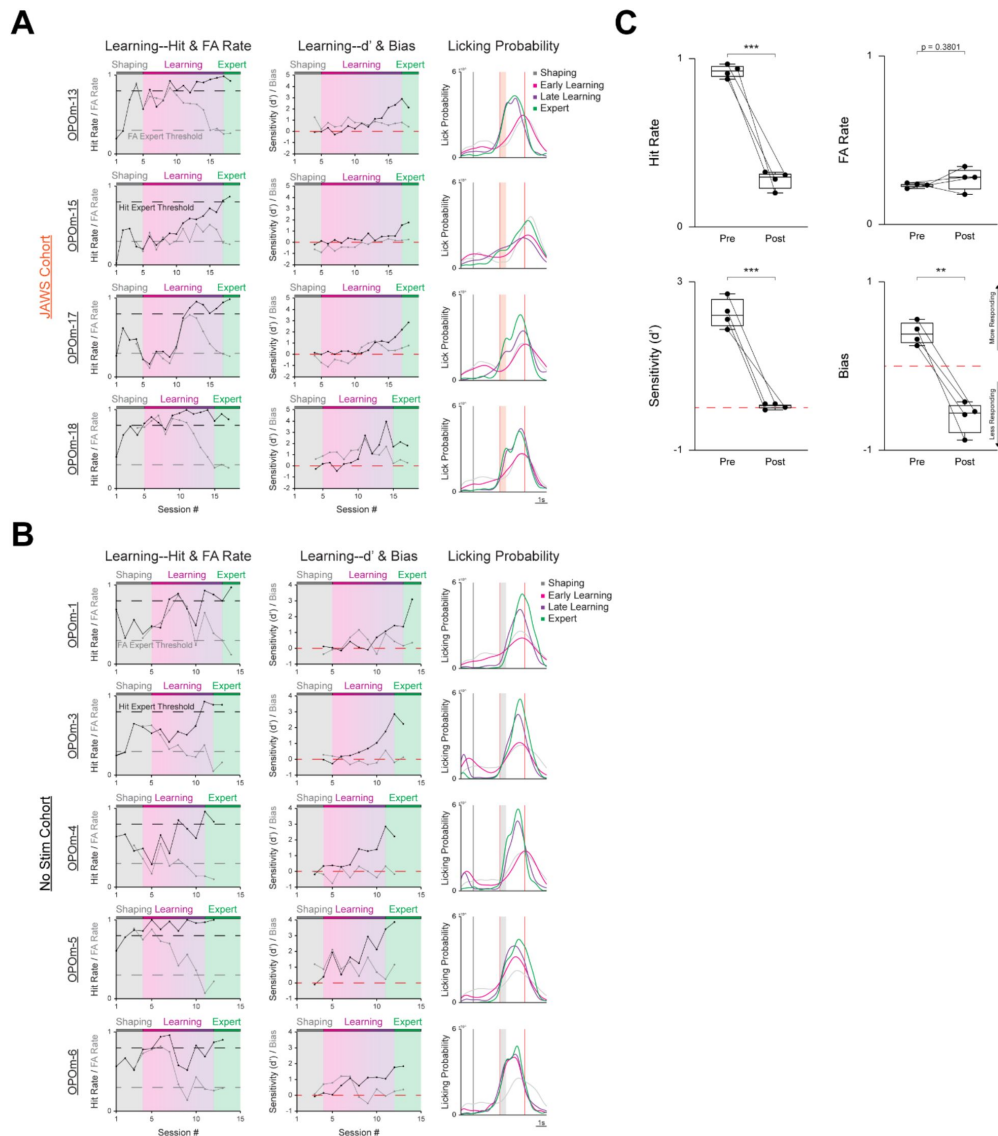


Figure S5.

Individual Longitudinal Learning-Related Changes in Behavioral Parameters and Licking Activity for the JAWS and No Stim cohorts, and Whisker Trim. Related to Figure 4

(A-B) Longitudinal learning-related changes as mice from the **(A)** JAWS and the No Stim **(B)** cohorts progress through each behavioral time point (Shaping, Early Learning, Late Learning, and Expert). The Shaping time point permitted mice to learn to lick the spout for a water reward during the first three sessions without the presence of the textures. During the last two Shaping sessions, the Go and NoGo textures were presented simultaneously with a Go texture probability of 0.90 for the first session and 0.75 for the second session. For the first Learning session, and all subsequent sessions, the Go texture probability was set to 0.50. Mice were considered Learning at this time point. The Early Learning time point was considered the first two sessions after Shaping, while the Late Learning was considered the last two sessions before achieving Expert status. Mice were considered Expert when they had a Hit Rate ≥ 0.80 and a FA Rate ≤ 0.30 for two consecutive sessions. *Left*: Hit Rate (black) and FA Rate (gray) across learning. *Middle*: Sensitivity (d' ; black) and Bias (gray) across learning. Red dashed line indicates 0. Note that a strict d' threshold was not used due to artificially increased d' values as Hit Rate and/or FA Rate approached their extremes (0 or 1). *Right*: Probability density function for licking activity at each time point. Scale = 1s.

(C) Following the attainment of expert status, a group of mice ($n = 4$) underwent a whisker trim session. In this session, mice performed 75 trials as normal with no optogenetic stimulation. After, the right whiskers contacting the textures were trimmed, and another 75 trials were performed. Behavioral performance, based on the four parameters (Hit Rate, FA Rate, Sensitivity, and Bias), was compared. Data are mean $\pm$ SEM. ** $p < 0.01$, *** $p < 0.001$.

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Reviewer #1 (Public review):

Summary:

This work aims at understanding the role of thalamus POM in dorsal lateral striatum (DLS) projection in learning a sensorimotor associative task. The authors first confirm that POM forms "en passant" synapses with some of the DLS neuronal subtypes. They then perform a go/no-go associative task that consists of the mouse learning to discriminate between two

different textures and to associate one of them with an action. During this task they either record the activity of the POM to DLS axons using endoscopy or silence their activity. They report that POM axons in the DLS are activated around the sensory stimulus but that the activity is not modulated by the reward. Last, they showed that silencing the POM axons at the level of DLS slows down learning the task.

The authors show convincing evidence of projections from POM to DLS and that POM inputs to DLS code for whisking whatever the outcome of the task is. However, their results do not allow to conclude if more neurones are recruited during the learning process or if the already activated fibres get activated more strongly. Last, because POM fibres in the DLS are also projecting to S1, silencing the POM fibres in the DLS could have affected inputs in S1 as well and therefore, the slowdown in acquiring the task is not necessarily specific to the POM to DLS pathway.

Strengths:

One of the main strengths of the paper is to go from slice electrophysiology to behaviour to get an in-depth characterization of one pathway. The authors did a comprehensive description of the POM projections to the DLS using transgenic mice to unambiguously identify the DLS neuronal population. They also used a carefully designed sensorimotor association task, and they exploited the results in depth.

It is a very nice effort to have measured the activity of the axons in the DLS not only after the mice have learned the task but throughout the learning process. It shows the progressive increase of activity of POM axons in the DLS, which could imply that there is a progressive strengthening of the pathway. The results show convincingly that POM axons in the DLS are not activated by the outcome of the task but by the whisker activity, and that this activity in average increases with learning.

Weaknesses:

One of the main targets of the striatum from thalamic input are the cholinergic neurons that weren't investigated here, is there information that could be provided?

It is interesting to know that the POM projects to all neuronal types in the DLS, but this information is not used further down the manuscript so the only take-home message of Figure 1 is that the axons that they image or silence in the DLS are indeed connected to DLS neurons and not just passing fibres. In this line, are these axons the same as the ones projecting to S1? If this is the case, why would we expect a different behaviour of the axon activity at the DLS level compared to S1?

The authors used endoscopy to measure the POM axons in the DLS activity, which makes it impossible to know if the progressive increase of POM response is due to an increase of activity from each individual neurons or if new neurons are progressively recruited in the process.

The picture presented in Figure 4 of the stimulation site is slightly concerning as there are hardly any fibres in neocortical layer 1 while there seems to be quite a lot of them in layer 4, suggesting that the animal here was injected in the VB. This is especially striking as the implantation and projection sites presented in Figure 1 and 2 are very clean and consistent with POM injection.

Comment after review: The weaknesses remain as concerns have not been addressed. The dataset is interesting but the interpretation, due partly to the lack of control (especially relative to VPM contamination), is difficult.

<https://doi.org/10.7554/eLife.98563.2.sa3>

Reviewer #2 (Public review):

Summary:

Yonk and colleagues show that the posterior medial thalamus (POm), which is interconnected with sensory and motor systems, projects directly to major categories of neurons in the striatum, including direct and indirect pathway MSNs, and PV interneurons. Activity in POm-striatal neurons during a sensory-based learning task indicates a relationship between reward expectation and arousal. Inhibition of these neurons slows reaction to stimuli and overall learning. This circuit is positioned to feed salient event activation to the striatum to set the stage for effective learning and action selection.

Strengths:

The results are well presented and offer interesting insight into an understudied thalamostriatal circuit. In general, this work is important as part of a general need for an increased understanding of thalamostriatal circuits in complex learning and action selection processes, which have generally received less attention than corticostriatal systems.

Weaknesses:

There could be a stronger connection between the connectivity part of the data - showing that POm neurons project to D1, D2, and PV neurons in striatum but with some different properties - and the functional side of the project. One wonders whether the POm neurons projecting to these subtypes or striatal neurons have unique signaling properties related to learning, or if there is a uniform, bulk signal sent to striatum. This is not a weakness per se, as it's reasonable for these questions to be answered in future papers.

All the in vivo activity-related conclusions stem from data from just 5 mice, which is a relatively small sample set. Optogenetic groups are also on the small side.

Comments on revisions:

The revision has a lot of thoughtful discussion added. I think overall the paper is more thorough and will also be a nice set up for a number of future research questions.

<https://doi.org/10.7554/eLife.98563.2.sa2>

Reviewer #3 (Public review):

Yonk and colleagues investigate the role of the thalamostriatal pathway. Specifically, they studied the interaction of the posterior thalamic nucleus (PO) and the dorsolateral striatum in the mouse. First, they characterize connectivity by recording DLS neurons in in vitro slices and optogenetically activating PO terminals. PO is observed to establish depressing synapses onto D1 and D2 spiny neurons as well as PV neurons. Second, the image PO axons are imaged by fiber photometry in mice trained to discriminate textures. Initially, no trial-locked activity is observed, but as the mice learn PO develops responses timed to the audio cue that marks the start of the trial and precedes touch. PO does appear to encode the tactile stimulus type or outcome. Optogenetic suppression of PO terminals in striatum slows task acquisition. The authors conclude that PO provides a "behaviorally relevant arousal-related signal" and that this signal "primes" striatal circuitry for sensory processing.

A great strength of this paper is its timeliness. Thalamostriatal processing has received almost no attention in the past, and the field has become very interested in the possible functions of PO. Additionally, the experiments exploit multiple cutting-edge techniques.

There seem to be some technical/analytical weaknesses. The in vitro experiments appear to have some contamination of nearby thalamic nuclei by the virus delivering the opsin, which could change the interpretation. Some of the statistical analysis of these data also appear inappropriate. The correlative analysis of Pom activity in vivo, licking, and pupil could be more convincingly done.

The bigger weakness is conceptual - why should striatal circuitry need "priming" by thalamus in order to process sensory stimuli? Why would such circuitry even be necessary? Why is a sensory signal from cortex insufficient? Why should the animal more slowly learn the task? How does this fit with existing ideas of striatal plasticity? It is unclear from the experiments that the thalamostriatal pathway exists for priming sensory processing. In fact the optogenetic suppression of the thalamostriatal pathway seems to speak against that idea.

Comments on revisions:

The authors have only tweaked the Discussion and not necessarily in ways that addressed our previous comments. They could have fairly easily analyzed the effect of distance of recording from injection site and compared subsets of data depending on contamination beyond PO (my comments 1 and 2) or effects of movements (3 and 4). Minimally, they could have given caveats in the Results and Discussion about these, and I would strongly encourage them to be explicit about the caveats. The analyses would probably be better.

The suggestion that the effects have something to do with priming (5), seems a grasp for function of the circuit.

<https://doi.org/10.7554/eLife.98563.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

This work aims to understand the role of thalamus POM in dorsal lateral striatum (DLS) projection in learning a sensorimotor associative task. The authors first confirm that POM forms "en passant" synapses with some of the DLS neuronal subtypes. They then perform a go/no-go associative task that consists of the mouse learning to discriminate between two different textures and to associate one of them with an action. During this task, they either record the activity of the POM to DLS axons using endoscopy or silence their activity. They report that POM axons in the DLS are activated around the sensory stimulus but that the activity is not modulated by the reward. Last, they showed that silencing the POM axons at the level of DLS slows down learning the task.

The authors show convincing evidence of projections from POM to DLS and that POM inputs to DLS code for whisking whatever the outcome of the task is. However, their results do not allow us to conclude if more neurons are recruited during the learning process or if the already activated fibres get activated more strongly. Last, because POM fibres in the DLS are also projecting to S1, silencing the POM fibres in the DLS could have affected inputs in S1 as well and therefore, the slowdown in acquiring the task is not necessarily specific to the POM to DLS pathway.

We thank the reviewer for these constructive comments. The points are addressed below.

Strengths:

One of the main strengths of the paper is to go from slice electrophysiology to behaviour to get an in-depth characterization of one pathway. The authors did a comprehensive description of the POM projections to the DLS using transgenic mice to unambiguously identify the DLS neuronal population. They also used a carefully designed sensorimotor association task, and they exploited the results in depth.

It is a very nice effort to have measured the activity of the axons in the DLS not only after the mice have learned the task but throughout the learning process. It shows the progressive increase of activity of POM axons in the DLS, which could imply that there is a progressive strengthening of the pathway. The results show convincingly that POM axons in the DLS are not activated by the outcome of the task but by the whisker activity, and that this activity on average increases with learning.

Weaknesses:

One of the main targets of the striatum from thalamic input are the cholinergic neurons that weren't investigated here, is there information that could be provided?

This is true of the parafascicular (Pf) thalamic nucleus, which has been well studied in this context. However, there is much less known about the striatal projections of other thalamic nuclei, including POM, and their inputs to cholinergic neurons. Anatomical tracing evidence from Klug et al. (2018), which mapped brain-wide inputs to striatal cholinergic (ChAT) interneurons, suggests that Pf provides the majority of thalamic innervation of striatal ChAT neurons compared to other thalamic nuclei. Many other thalamic nuclei, including POM, showed very little of no labeling, suggesting weak innervation of ChAT interneurons. However, it is possible that these thalamic nuclei, including POM, do provide functional innervation of ChAT interneurons that is not sufficiently assessed by anatomical tracing. Understanding the innervation patterns of POM-striatal projections beyond the three cell types we have studied here would be an important area of further study.

It is interesting to know that the POM projects to all neuronal types in the DLS, but this information is not used further down the manuscript so the only take-home message of Figure 1 is that the axons that they image or silence in the DLS are indeed connected to DLS neurons and not just passing fibres. In this line, are these axons the same as the ones projecting to S1? If this is the case, why would we expect a different behaviour of the axon activity at the DLS level compared to S1?

Tracing of single POM axons by Ohno et al. (2012) indicated that POM axons form a branched collateral that innervates striatum, while the main axon continues in the rostral-dorsal direction to innervate cortex. We think it is reasonable, based on the morphology, that our optogenetic suppression experiment restricted the suppression of glutamate release to this branch and avoided the other branches of the axon that project to cortex. However, testing this would require monitoring S1 activity during the POM-striatal axon suppression, which we did not do in this study.

It is a very interesting question whether there could be different axon activity behavior in striatum versus S1. There is surprising evidence that POM synaptic terminals are different sizes in S1 and M1 and show different synaptic physiological properties depending on these cortical projection targets (Casas-Torremocha et al., 2022). Based on this, it is possible that POM-striatal synapses show distinct properties compared to cortex; however, this will need to be tested in future work.

The authors used endoscopy to measure the POM axons in the DLS activity, which makes it impossible to know if the progressive increase of POM response is due to an increase of activity from each individual neuron or if new neurons are progressively recruited in the process.

This is a good point. It would be necessary to perform chronic two-photon imaging of POM neurons (or chronic electrophysiological recordings) to determine whether the activity of individual neurons increased versus whether individual neuron activity levels remained similar but new neurons became active with learning. Even under baseline conditions, it is not known in detail what fraction of the population of POM neurons is active during sensory processing or behavior, highlighting how much is still to be discovered in this exciting area of neuroscience.

The picture presented in Figure 4 of the stimulation site is slightly concerning as there are hardly any fibres in neocortical layer 1 while there seems to be quite a lot of them in layer 4, suggesting that the animal here was injected in the VB. This is especially striking as the implantation and projection sites presented in Figures 1 and 2 are very clean and consistent with POM injection.

Although this image was selected to demonstrate the position of the POM injection site and optical fiber implant above striatal axons, the reviewer is correct that there appears to be mixed labeling of axons in L4 and L5a. In some cases, there was expression slightly outside the border of POM (see Fig. 1B, right), which might explain the cortical innervation pattern in this figure. While cortically bound VPM axons pass through the striatum, they do not form synaptic terminals until reaching the cortex (Hunnicutt et al., 2016). If, as may be the case, inhibitory opsins suppress release of neurotransmitter at synaptic terminals more effectively than action potential propagation in axons, it may be likely that optogenetic suppression of POM-striatal terminals is more effective than suppression of action potentials in off-target-labelled VPM axons of passage. Ideally, we could compare effects of suppression of POM-striatal synapses with POM-cortical synapses and VPM-cortical synapses, but this was outside the bandwidth of the present study.

Reviewer #2 (Public Review):

Summary:

Yonk and colleagues show that the posterior medial thalamus (POM), which is interconnected with sensory and motor systems, projects directly to major categories of neurons in the striatum, including direct and indirect pathway MSNs, and PV interneurons. Activity in POM-striatal neurons during a sensory-based learning task indicates a relationship between reward expectation and arousal. Inhibition of these neurons slows reaction to stimuli and overall learning. This circuit is positioned to feed salient event activation to the striatum to set the stage for effective learning and action selection.

Strengths:

The results are well presented and offer interesting insight into an understudied thalamostriatal circuit. In general, this work is important as part of a general need for an increased understanding of thalamostriatal circuits in complex learning and action selection processes, which have generally received less attention than corticostriatal systems.

Weaknesses:

There could be a stronger connection between the connectivity part of the data - showing that POM neurons context D1, D2, and PV neurons in the striatum but with some different properties - and the functional side of the project. One wonders whether the POM neurons projecting to these subtypes or striatal neurons have unique signaling properties related to learning, or if there is a uniform, bulk signal sent to the striatum. This is not a weakness per se, as it's reasonable for these questions to be answered in future papers.

We are very interested to understand the potentially distinct learning-related synaptic and circuit changes that potentially occur at the POM synapses with D1- and D2-SPNs and PV interneurons, and other striatal cell types. We agree that this would be an important topic for further investigation.

All the in vivo activity-related conclusions stem from data from just 5 mice, which is a relatively small sample set. Optogenetic groups are also on the small side.

We appreciate this point and agree that higher N can be important for observing robust effects. A factor of our experiments that helped reduce the number of animals used was the longitudinal design, with repeated measures in the same subjects. This allowed for the internal control of comparing learning effects in the same subject from naïve to expert stages and therefore increased robustness. Even with relatively small group sizes, results were statistically significant, suggesting that the use of more mice was unnecessary, which we considered consistent with best practice in the use of animals in research. We also note that our group sizes were consistent with other studies in the field.

Reviewer #3 (Public Review):

Yonk and colleagues investigate the role of the thalamostriatal pathway. Specifically, they studied the interaction of the posterior thalamic nucleus (PO) and the dorsolateral striatum in the mouse. First, they characterize connectivity by recording DLS neurons in in-vitro slices and optogenetically activating PO terminals. PO is observed to establish depressing synapses onto D1 and D2 spiny neurons as well as PV neurons. Second, the image PO axons are imaged by fiber photometry in mice trained to discriminate textures. Initially, no trial-locked activity is observed, but as the mice learn PO develops responses timed to the audio cue that marks the start of the trial and precedes touch. PO does appear to encode the tactile stimulus type or outcome. Optogenetic suppression of PO terminals in striatum slow task acquisition. The authors conclude that PO provides a "behaviorally relevant arousal-related signal" and that this signal "primes" striatal circuitry for sensory processing.

A great strength of this paper is its timeliness. Thalamostriatal processing has received almost no attention in the past, and the field has become very interested in the possible functions of PO. Additionally, the experiments exploit multiple cutting-edge techniques.

There seem to be some technical/analytical weaknesses. The in vitro experiments appear to have some contamination of nearby thalamic nuclei by the virus delivering the opsin, which could change the interpretation. Some of the statistical analyses of these data also appear inappropriate. The correlative analysis of Pom activity in vivo, licking, and pupil could be more convincingly done.

The bigger weakness is conceptual - why should striatal circuitry need "priming" by the thalamus in order to process sensory stimuli? Why would such circuitry even be necessary? Why is a sensory signal from the cortex insufficient? Why should the animal more slowly learn the task? How does this fit with existing ideas of striatal plasticity? It is unclear from the experiments that the thalamostriatal pathway exists for priming

sensory processing. In fact, the optogenetic suppression of the thalamostriatal pathway seems to speak against that idea.

We thank the reviewer for these constructive comments. The points are addressed below.

Recommendations for the authors:

Reviewer #2 (Recommendations For The Authors):

Do POM neurons innervate CINs also? The connection between the PF thalamus and CINs is mentioned in a couple of places - one question is how unique are the input patterns for the POM versus adjacent sensorimotor thalamic regions, including the PF? This isn't a weakness per se but knowing the answer to that question would help in forming a more complete picture of how these different thalamostriatal circuits do or do not contribute uniquely to learning and action selection.

Anatomical tracing evidence from Klug et al. (2018), which mapped brain-wide inputs to striatal cholinergic (ChAT) interneurons, suggests that Pf provides the majority of thalamic innervation of striatal ChAT neurons compared to other thalamic nuclei. Many other thalamic nuclei, including POM, showed very little or no labeling, suggesting weak innervation of ChAT interneurons. However, it is possible that these thalamic nuclei, including POM, do provide functional innervation of ChAT interneurons that is not sufficiently assessed by anatomical tracing.

Another difference between Pf and other thalamic nuclei (likely including POM) comes from anatomical tracing evidence (Smith et al., 2014; PMID: 24523677) which indicates that Pf inputs form the majority of their synapses onto dendritic shafts of SPNs, while other thalamic nuclei form synapses onto dendritic spines. Understanding the innervation patterns of POM-striatal projections beyond the three cell types we have studied here, including ChAT neurons and subcellular localization, would be an important area of further study.

It would be useful to know to what extent these POM-striatum neurons are activated generally during movement, versus this discrimination task specifically.

We agree that distinguishing general movement-related activity from task-specific activity would be very useful. Earlier work (Petty et al., 2021) showed a close relationship between POM neuron activity, spontaneous (task-free) whisker movements, and pupil-indexed arousal in head-restrained mice. Oram et al. (2024; PMID: 39003286) recently recorded VPM and POM in freely moving mice during natural movements, finding that activity of both nuclei correlated with head and whisker movements. These studies indicate that POM is generally coactive with exploratory head and whisker movements.

During task performance, the situation may change with training and attentional effects. For example, Petty and Bruno (2024) (<https://elifesciences.org/reviewed-preprints/97188>) showed that POM activity correlates more closely with task demands than tactile or visual stimulus modality. Our data indicate that POM axonal signals are increased at trial start during anticipation of tactile stimulus delivery and through the sensory discrimination period, then decrease to baseline levels during licking and water reward collection (Fig. 3). Results of Petty and Bruno (2024) together with ours suggest that POM is particularly active during the context of behaviorally relevant task performance. Thus, we think it is likely that, while pupil dilation indexes general movement and arousal, POM activity is more specific to movement and arousal associated with task engagement and behavioral performance. We have strengthened this point in the Discussion.

Many of the data panels and text for legends/axes are quite small, and the stroke on line art is quite faint - overall figures could be improved from a readability standpoint.

We thank the reviewer for their careful attention to the figures.

Reviewer #3 (Recommendations For The Authors):

Major

(1) Page 4, the Results regarding PSP and distance from injection site. The r-squared is the wrong thing to look at to test for a relationship. One should look at the p-value on the coefficient corresponding to the slope. The p-value is probably significant given the figures, in which case there may be a relationship contrary to what is stated. All the low r-squared value says is that, if there is a relationship, it does not explain a lot of the PSP variability.

We thank the reviewer for alerting us this oversight. We have included the p value ($p = 0.0293$) in the figure and legend, and indicated that the relationship is “small but significant”.

(2) Figure 1B suggests that the virus injections extend beyond POM and into other thalamic structures. Do any of the results change if the injections contaminating other nuclei are excluded from the analysis? I am not suggesting the authors change the figures/analyses. I am simply suggesting they double-check.

We selected for injections that were predominantly expressing in POM as determined by post-hoc histological analysis (see Fig. 1, right). As above, we think that axons of passage that do not form striatal synapses are less likely to be suppressed than axons with terminals; however, this would need to be determined in further experiments. Because the preponderance of expression is within POM, we think the results would be similar even with a stricter selection criterion.

(3) The authors conclude that POM and licking are not correlated (bottom of page 6 pertaining to Figures 3A-F). The danger of these analyses is that they assume that GCaMP8 is a perfect linear reporter of POM spikes. The reliability of GCaMP8 has been quantified in some cell types, but not thalamic neurons, which have relatively higher firing rates.

The reviewer is correct that the relationship between GCaMP8 fluorescence changes and spiking has not been sufficiently characterized in thalamic neurons, and that this would be important to do.

What if the indicator is simply saturated late into the trial (after the average reaction time)? It would look like there is no response and one would conclude no correlation, but there could be a very strong correlation.

While saturation is worthy of concern, the signal dynamics here argue against this possibility. The reason is that the signal increased in the early part of the trial and decreased by the end. If saturation was an issue, this would have been apparent during the initial increase. When the signal decreased in amplitude at the end of the trial, this indicates that the signal is not saturated because it is returning from a point closer to its maximum (and is becoming less saturated).

Also, what happens between trials? Are the correlations the same, stronger, weaker? Ideally, the authors would analyze the data during and between trials.

Between trials the signal did not show further changes in baseline beyond what was displayed at the start and end of behavioral trials. There were no consistent increases or decreases in signals between trials, except perhaps during strong whisking bouts. This is anecdotal because we did not analyze between-trial data. However, it is interesting and important to note that signals increased dramatically in amplitude from naïve, early learning to expert behavioral performance (Fig. 3), highlighting that POM-axonal signals relate to behavioral engagement and performance rather than spontaneous behaviors.

(4) Axonal activity could also appear more correlated with the pupil than licking because pupil dynamics are slow like the dynamics of calcium indicators. These kernels could artificially inflate the correlation. Ideally, the authors could consider these temporal effects. Perhaps they could deconvolve the temporal profiles of calcium and pupil before correlating? Or equivalently incorporate the profiles into their analysis?

We analyzed the lick probability histograms, which had a temporal profile similar to the calcium signals (Fig. 3D,E), ruling out concerns about effects of temporal effects on correlations. It is also worth noting that we observed changes in correlations between calcium signals and pupil with learning stage (Fig. 3I), even though the temporal profiles (signal dynamics) are not changing. Thus, temporal effects of the signals themselves are not the driver of correlations, but rather the changes in relative timing between calcium signals and pupil, as occur with learning.

(5) The authors conclude that PO provides a "behaviorally relevant arousal-related signal" and that this signal "primes" striatal circuitry for sensory processing. The data here support the first part. It is not clear that the data support the second part, largely because it is vague what "priming" of sensory processing or "a key role in the initial stages of action selection (p.9) even means here. Why would such circuitry even be necessary? Why is a sensory signal from the cortex insufficient? Why should the animal more slowly learn the task? How does this fit with existing ideas of striatal plasticity? Some conceptual proposals from the authors, even if speculative and not offered as a conclusion, would be helpful.

We appreciate these good points and have added further consideration and revision of the concept of priming and potential roles in an extensively revised Discussion section.

(6) The photometry shows that PO turns on about 2 seconds before the texture presentation. PO's activity seems locked to the auditory cue, not the texture (Figure 2). This means that the attempt to suppress the thalamostriatal pathway with JAWS (Figure 4) is rather late, isn't it? Some PO signals surely go through. This seems to contradict the idea of priming above. It would be good if the authors could factor this into their narrative. Perhaps labelling the time of the auditory cue in Figure 4C would also be helpful.

The start of texture presentation (movement of the texture panel toward the mouse) and auditory cue occur at the same time. To clarify this, we added a label "start tone" in Figure 4C and also in Figure 2C.

For optogenetic (JAWS) suppression, we intentionally chose a time window between start tone onset and texture presentation, because our photometry experiments showed that this was when the preponderance of the signal occurred. However, the reviewer is correct that our chosen optogenetic suppression (JAWS) onset occurs shortly after the photometry signal has already started, potentially leaving the early photometry signal un-suppressed. Our motivation for choosing a restricted time window surrounding the texture presentation time was 1) to minimize illumination and potential heating of brain tissue; 2) to target a time

window that avoids the auditory cue but covers stimulus presentation. We did not want to extend the duration of the suppression to before the trial started, because this could produce task-non-specific effects, such as distraction or loss of attention before the start of the trial.

Even if some signal were getting through before suppression, we don't think this contradicts the possibility of 'priming', because the process underlying priming would still be disrupted even if not totally suppressed. This would alter the temporal relationship between POM-striatal inputs and further corticostriatal inputs (from S1 and M1 cortex, for example). We have included further consideration of these points and possible relation to the priming concept in the Discussion.

Minor

(1) Page 5, "the sensitivity metric is artificially increased". What do you mean "artificially"? The mice are discriminating better. It is true that either a change in HR or FAR can cause the sensitivity metric to change, but there is nothing artificial or misleading about this.

We removed the word artificial and clarified our definition of behaviorally Expert in this context:

"Mice were considered Expert once they had reached ≥ 0.80 Hit Rate and ≤ 0.30 FA Rate for two consecutive sessions in lieu of a strict sensitivity (d') threshold; we found this definition more intuitive because d' is enhanced as Hit Rate and FA Rate approach their extremes (0 or 1)"

(2) Page 7, "Upon segmentation (Figure S4G-J)". Do you mean "segregation by trial outcome"?

Corrected.

(3) Page 9, "POM projections may have discrete target-specific functions, such that POM-striatal inputs may play a distinct role in sensorimotor behavior compared to POM-cortical inputs". Would POM-cortical inputs not also be sensorimotor? The somatosensory cortex contains a lot of corticostriatal cells. It also has various direct and indirect links to the motor cortex as well.

We have clarified the wording here to convey the possibility that POM signals could be received and processed differently by striatal versus cortical circuitry, and have moved this statement to later in the discussion for better elaboration.

(4) The Methods state that male and female mice were used. Why not say how many of each and whether or not there are any sex-specific differences?

We added the following information to the Methods:

The number of male and female mice were as follows, by experiment type: 6 male, 4 female (electrophysiology); 3 male, 2 female (fiber photometry); 4 male, 5 female (optogenetics). Data were not analyzed for sex differences.

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